High-dimensional geometry of population responses in visual cortex

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A neuronal population encodes information most efficiently when its stimulus responses are high-dimensional and uncorrelated, and most robustly when they are correlated and lower-dimensional. Here, we analyzed the dimensionality of the encoding of natural images by large visual cortical populations recorded from awake mice. Evoked population activity was high dimensional, with correlations obeying an unexpected power-law: the n^{th} principal component variance scaled as $1/n$. This scaling was not inherited from the $1/f$ spectrum of natural images, because it persisted after stimulus whitening. We proved mathematically that if the variance spectrum decayed any slower, the population code could not be smooth, allowing small changes in input to dominate population activity. The theory also predicts larger power-law exponents for lower-dimensional stimulus ensembles, which we validated experimentally. These results suggest that coding smoothness may represent a fundamental constraint governing correlations in neural population codes.

Introduction

The visual cortex contains millions of neurons, and the patterns of activity that images evoke in these neurons form a "population code". The structure of this code is largely unknown, due to the lack of techniques able to record from large populations. Nonetheless, the population code is the subject of long-standing theories.

One such theory is the "efficient coding hypothesis" [1, 2, 3], which maintains that the neural code maximizes information transmission by eliminating correlations in natural image inputs. Such codes are high-dimensional and sparse, which can allow complex features to be read out by simple downstream networks [4, 5, 6].

However, several studies have suggested that neural codes are confined to low-dimensional subspaces ("planes") [7, 8, 9, 10, 11, 12, 13, 14, 15]. Codes of low planar dimension are correlated and redundant, allowing for robust computations of stimuli in the face of noise [16, 17]. Nevertheless, low planar dimension is inevitable given stimuli or tasks of limited complexity [18]: the responses to a set of n stimuli, for example, have to lie in an n dimensional subspace. The planar dimension of the cortical code thus remains an open question, which can only be answered by recording the responses of large numbers of neurons to large numbers of stimuli.

Here, we recorded the simultaneous activity of $\sim$10,000 neurons in mouse visual cortex, in response to thousands of natural images. We found that stimulus responses were neither uncorrelated ("efficient coding") nor low-dimensional. Instead, responses occupied a multidimensional space with the variance in the n^{th} dimension scaling as a power law $n^{-\alpha}$, where $\alpha \approx 1$. We showed mathematically that if variances decay slower than a power law with exponent $\alpha = 1 + 2/d$, where d is the dimension of the input ensemble, then the space of neural activity must be non-differentiable – i.e. not smooth. We varied the dimensionality of the stimuli d and found that the neural responses respected this lower bound. These findings suggest that the population responses are constrained by efficiency, to make best use of limited numbers of neurons, and smoothness, which allows similar images to evoke similar responses.

Simultaneous recordings of $\sim$10,000 neurons

To obtain simultaneous recordings of $\sim$10,000 cells from mouse V1, we employed resonance-scanning two-photon calcium microscopy, using 11 imaging planes spaced at 35μm (Fig. 1a). The slow timecourse of the GCaMP6s sensor allowed activity to be detected at a 2.5 Hz scan rate, and an efficient data processing pipeline [19] allowed large numbers of cells to be detected accurately (Fig. 1b). Natural image scenes (Imagenet database [20]) were presented on an array of 3 monitors surrounding the mouse (Fig. 1c), at an average of 1 image/s. Cells were tuned to these natural image stimuli: in experiments in which

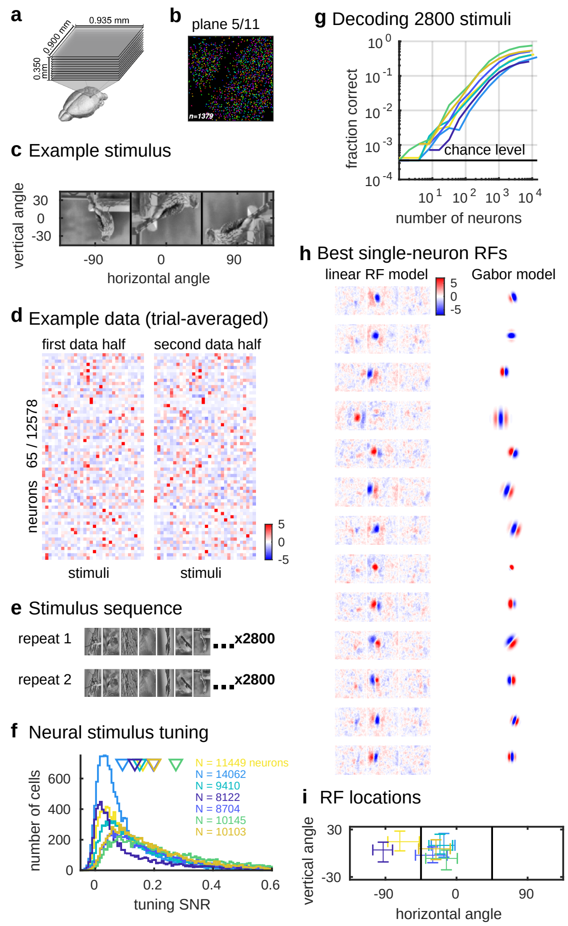

Figure 1: **Population coding of visual stimuli.** **a**, Simultaneous recording of ~10,000 neurons using 11-plane two-photon calcium imaging. **b**, Randomly-pseudocolored cells in an example imaging plane. **c**, Example stimulus spans three screens surrounding the mouse's head. **d**, Mean responses of 65 randomly-chosen neurons to 32 image stimuli (96 repeats, z-scored, scale bar represents standard deviations, one recording out of four shown). **e**, A sequence of 2800 stimuli was repeated twice during the recording. **f**, Distribution of single-cell signal-to-noise ratios (SNR) (2800 stimuli, two repeats). Colors denote recordings; arrows represent means. **g**, Stimulus decoding accuracy as a function of neuron count for each recording. **h**, Example receptive fields (RFs) fit using reduced-rank regression or Gabor models (z-scored) (one recording shown, out of 7). **i**, Distribution of the receptive field centers, plotted on the left and center screens (line denotes screen boundary). Each cross represents a different recording, with 95% of neuron's RF centers within error bars.

responses to 32 images were averaged over 96 repeats (Fig. 1d), stimulus responses accounted for 55.4±3.3% (SE, n=4 recordings) of the trial-averaged variance. Consistent with prior reports [21, 22, 23], neuronal responses were sparse: only a small fraction of cells (13.4±1.0% SE, n=4 recordings) were driven more than two standard deviations above their baseline firing rate by any particular stimulus.

For our main experiments, we assembled a sequence of 2,800 image stimuli, and presented these stimuli twice in the same order, to maximize the number of images presented while still allowing analyses based on cross-validation (Fig. 1e). Most neurons (81.4±5.1% SE, n=7 recordings) showed correlation between repeats at $p < 0.05$ (Extended Data Fig. 1a,b). Nevertheless, consistent with previous reports [24], responses showed substantial trial-to-trial variability. Cross-validation showed that stimulus responses accounted for on average 13.2±1.5% of the single-trial variance (Extended Data Fig. 1c), and the average signal-to-noise ratio was 17.3±2.4% (Fig. 1f). This level of trial-to-trial variability was not due to our particular recording method: measuring responses to the same stimuli electrophysiologically yielded a similar signal-to-noise ratio (Extended Data Fig. 2). Despite trial-to-trial variability, however, population activity recorded on a single trial contained substantial information about the sensory stimuli. A simple nearest-neighbor decoder, trained on one repeat and tested on the other, was able to identify the presented stimulus with up to 75.5% accuracy (Fig. 1g; range 25.4%-75.5%; median 41.7% compared to chance level of 0.036%, n=7 recordings). Decoding accuracy did not fully saturate at a population size of 10,000, suggesting that performance would increase further with more neurons.

Neurons had similar visual properties to previous reports [23, 25], and their responses were only partially captured by classical linear-nonlinear models, consistent with previous visual cortex studies [26, 27, 28, 29, 30]. We calculated a receptive field (RF) for each cell from its responses to natural images in two ways: by fitting linear RFs regularized with a reduced rank method; or by searching for an optimal Gabor filter that was rectified/quadrature filtered to simulate classical simple/complex cell responses. As expected from retinotopy, the RF locations of simultaneously recorded neurons overlapped but there was a high diversity of receptive field sizes and shapes (Fig. 1h; Extended Data Fig. 3). Both RF models, however, explained less than 20% of the stimulus-related variance (the linear model explained 11.4±0.7% SE, and the Gabor model explained 18.5±1.0% SE, n=7 recordings each).

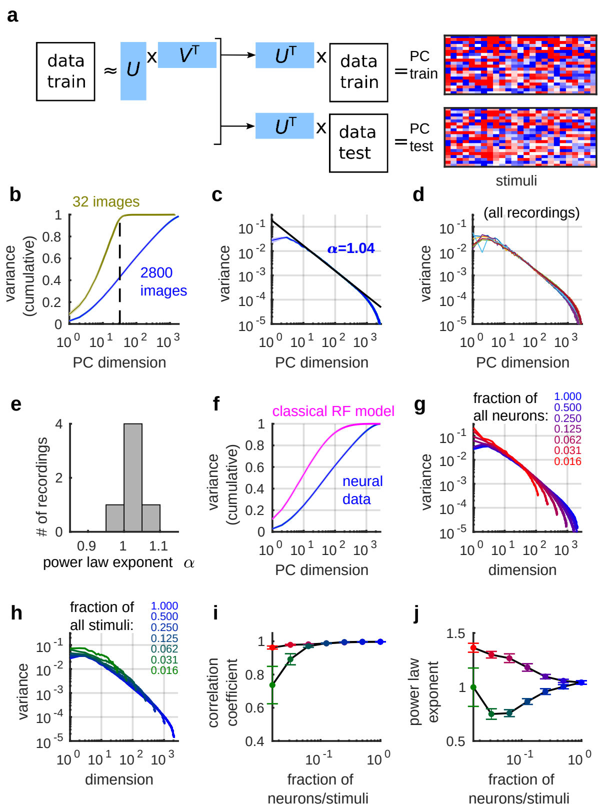

Figure 2: **Visual cortical responses are high-dimensional with power-law eigenspectrum.** **a**, The eigenspectrum of visual stimulus responses was estimated by cross-validated principal component analysis (cvPCA), projecting singular vectors from the first repeat onto responses from the second. **b**, Cumulative fraction of variance in planes of increasing dimension, for an ensemble of 2800 stimuli (blue) and for 96 repeats of 32 stimuli. Dashed line indicates 32 dimensions. **c**, Eigenspectrum plotted in descending order of training set singular value for each dimension, averaged across 7 recordings (shaded error bars represent standard error). Black line denotes linear fit of $1/n^{\alpha}$. **d** Eigenspectra of each recording individually. **e**, Histogram of power law exponents α across all recordings. **f**, Cumulative eigenspectrum for a simple/complex Gabor model fit to the data (pink) superimposed on true data (blue). **g**, Eigenspectra computed from random subsets of recorded neurons, fraction indicated by colors. **h**, Same analysis for random subsets of stimuli. **i**, Pearson correlation of log variance and log dimension over dims 11-500, as a function of fraction analyzed (1 indicates a power law). **j**, Power law exponents of the spectra plotted in **g**,**h**.

Power-law scaling of dimensionality

To characterize the geometry of the population code for visual stimuli, we developed a method of cross-validated principal component analysis (cvPCA). cvPCA measures the reliable variance of stimulus-related dimensions, excluding trial-to-trial variability from unrelated cognitive/behavioral variables or noise. It accomplishes this by computing the covariance of responses between training and test presentations of an identical stimulus ensemble (Fig. 2a). Because only stimulus-related activity will be correlated across repeats, cvPCA provides an unbiased estimate of the stimulus-related variance. In simulations using the same noise statistics as our recordings, we confirmed that this technique can recover the true variances (Extended Data Fig. 4, Extended Data Fig. 5, Supplementary Discussion 1).

This method revealed that the visual population responses did not lie on any low-dimensional plane within the space of possible firing patterns. The amount of variance explained continued to increase as further dimensions were included, without saturating at any dimensionality below the maximum possible (Fig. 2b). As a control analysis, we applied cvPCA to neural responses to only 32 images shown many times – whose responses must by definition lie in a 32-dimensional subspace – and indeed observed a saturation of the variance after 32 dimensions.

This analysis revealed an unexpected finding: the fraction of neural variance in planes of successively larger dimensions followed a power law. The eigenspectrum, i.e. the function summarizing the variance of the n^{th} principal component, had a magnitude approximately proportional to $1/n$ (Fig. 2c), reflecting successively less variance in dimensions encoding finer stimulus features (Extended Data Fig. 6). Its power-law structure did not result from averaging over experiments: analysis of data from each mouse individually revealed power-law behavior in every case (Fig. 2d). The scaling exponent of the power law had a peak just above 1 (1.04±0.02 SE, n = 7 recordings, Fig. 2e). This eigenspectrum reflected correlations between neurons, and was not the consequence of a log-normal distribution of firing rates or signal variance (Extended Data Fig. 7). In addition, this result could not be explained by classical models of visual cortical receptive fields: a model of simple/complex Gabor receptive fields with parameters fit to single cell responses (Fig. 1h) had lower dimensionality than the neural responses (Fig. 2f).

The range of dimensions over which the power law held grew larger the more neurons and stimuli were considered. To show this, we repeated the analyses on randomly-

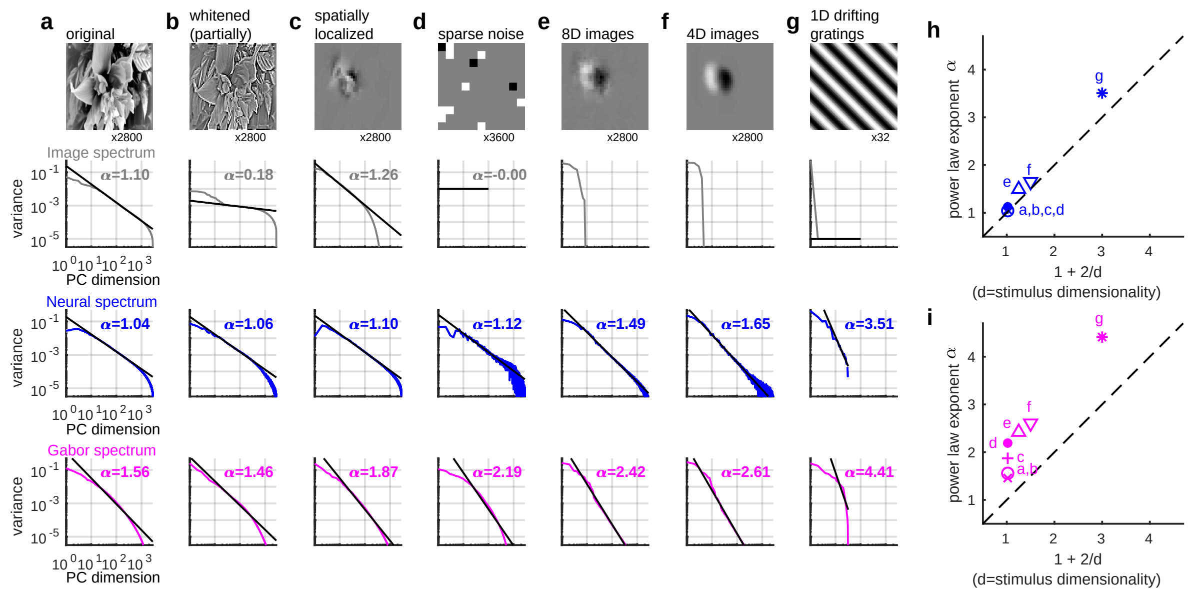

Figure 3: **Power law exponent depends on input dimensionality, but not image statistics.** **a**, Eigenspectra of natural image pixel intensities (gray), of visual cortical responses to these stimuli (blue), and of a simple/complex cell model's response to these stimuli (pink). **b**, Same analysis for responses to spatially whitened stimuli, that lack $1/n$ image spectrum. **c**, Same analysis for images windowed over the RF of the recorded population. **d**, Same analysis for sparse noise stimuli. **e**, Same analysis for images projected into 8 dimensions produces a faster eigenspectrum decay with exponent α=1.49. **f**, After projecting images to 4 dimensions, α=1.65. **g**, Responses to drifting gratings, a one dimensional stimulus ensemble, show yet faster decay with α=3.43. **h,i**, Summary of power law exponents α for neural responses (h) and Gabor model (i), as a function of the dimensionality of the stimulus set d. Dashed line: $\alpha = 1 + 2/d$, corresponding to the border of fractality.

chosen subsets of neurons or stimuli (Fig. 2g-h). To quantify the accuracy of the power law, we fit the log variance of the n^{th} dimension to $log(n)$ using linear regression over the range 11-500 dimensions. Both the correlation coefficient and slope (which reflects the power law exponent) approached 1 for increasing subset sizes (Fig. 2i,j, Extended Data Fig. 8). Electrophysiological recordings – with fewer recorded neurons and fewer presented stimuli – showed the same eigenspectrum as a similarly-sized subset of the two-photon data (Extended Data Fig. 9). We conclude that the power law held accurately over ~ 2 orders of mangitude in these recordings, and infer it would likely extend ever further if more yet neurons and stimuli could be analyzed.

Power-law and stimulus statistics

Natural images have a power-law structure[31, 32] (Fig. 3a), but this did not cause the neural code's power-law. To show this, we removed the image power law by spatially whitening the images, and presented the whitened stimuli to 3 of the mice. Although the power law in the image pixels was abolished, the power law in the neural responses remained (Fig. 3b). Furthermore, the eigenspectrum of neural responses could not be explained by straightforward receptive field properties: the responses of a simple/complex Gabor model produced eigenspectra that decayed more quickly than the actual responses, and were worse fit by a power-law (p<10^{-3}, Wilcoxon rank-sum test on Pearson correlations, Fig. 3a,b).

The power law eigenspectra also did not arise from other characteristics of natural images. To investigate the role of long-range image correlations, we constructed spatially localized image stimuli, in which the region outside the classical RF was replaced by gray. Again, the power law persisted with exponent close to 1 (Fig. 3c). Finally, we removed any natural image structure and showed sparse noise stimuli to the mice (Fig. 3d). Again, we observed a power-law spectrum, with exponent close to 1, although slightly larger than for the natural image stimuli (1.13±0.044 SE, n=3 recordings; $p = 0.067$, Wilcoxon two-sided rank sum test). We conclude that power-law spectra do not reflect neural processing of a property specific to natural images.

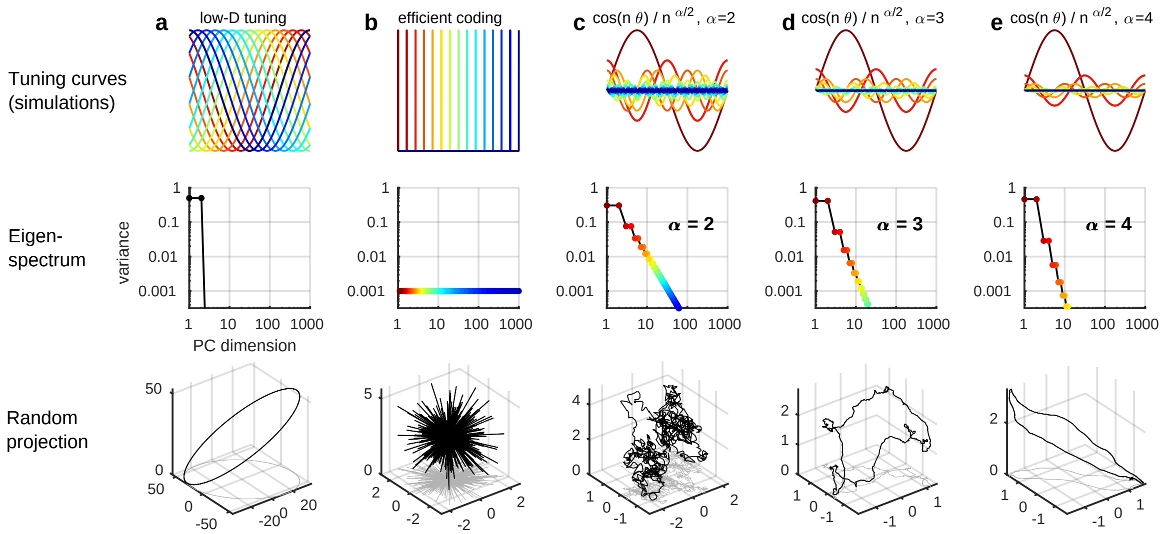

Figure 4: **The smoothness of simulated neural activity depends on the eigenspectrum decay.** Simulations of neuronal population responses to a 1-dimensional stimulus (x-axis), their eigenspectra, and a random projection of responses in 3D space. **a**, Wide tuning curves, corresponding to a circular neural manifold in a 2-dimensional plane. **b**, Narrow tuning curves corresponding to uncorrelated responses as predicted by the efficient coding hypothesis. **c-e**, Scale-free tuning curves corresponding to power law variance spectra, with exponents of 2, 3 (the critical value for $d = 1$), or 4. Tuning curves in c-e represent PC dimensions rather than individual simulated neurons.

Power-law and stimulus dimensionality

Power law eigenspectra are observed in many scientific domains, and are related to the smoothness of the underlying functions. For example, the Fourier spectrum of a differentiable function of one variable must decay at least as fast as a power law of exponent 1 (see e.g. Ref. [33]). We therefore theorized that the variance power law might be related to smoothness of the neural responses. We showed mathematically that if the sensory stimuli presented can be characterized by d parameters, and if the mapping from these parameters to (noise-free) neural population responses is differentiable, then the population eigenspectrum must decay asymptotically faster than a power law of exponent $\alpha = 1 + 2/d$ (Supplementary Discussion §2). Conversely, if the eigenspectrum decays slower that this, a smooth neural code is impossible as the neural responses must lie on a fractal rather than a differentiable manifold.

This mathematical analysis gave rise to an experimental prediction. For a high-dimensional stimulus ensemble such as natural images, d will be large so $1 + 2/d \approx 1$, close to the exponent we observed. However for smaller values of d, the power-law must have larger exponents if fractality is to be avoided. We therefore hypothesized that lower-dimensional stimulus sets would evoke population responses with larger power-law exponents.

We obtained stimulus ensembles of dimensionality $d = 8$ and $d = 4$ by projecting the natural images onto a set of d basis functions (Fig. 3e,f). For a stimulus ensemble of dimensionality $d = 1$ we used drifting gratings, parameterized by a single direction variable. Consistent with the hypothesis, stimulus sets with d = 8, 4, and 1 yielded power-law scaling of eigenvalues with exponents of 1.49, 1.65, and 3.43, near but above the lower bounds of 1.25, 1.50, and 3.00 predicted by the $1 + 2/d$ exponent (Fig. 3h). These results suggest that the neural responses lie on a manifold that is almost as high-dimensional as possible without becoming fractal. The eigenspectra of responses simulated from the simple/complex cell model decayed even faster, suggesting a differentiable, but lower dimensional representation (Fig. 3i).

Discussion

We found that the variance of n^{th} dimension of visual cortical population activity decays as a power of n, with exponent $\alpha \approx 1 + 2/d$ where d is the dimensionality of the space of sensory inputs. The population eigenspectrum reflects the fraction of neural variance devoted to representing coarse and fine stimulus features (Supplementary Discussion §2, §3; Extended Data Fig. 6). If the eigenspectrum decayed slower than $n^{-1-2/d}$ then the neural code would emphasize fine stimulus features so strongly that it could not be differentiable. Our results therefore suggest that the eigenspectrum of the visual cortical code decays almost as slowly as possible consistent with smooth neural coding.

To illustrate the consequences of eigenspectrum decay for neural codes, we simulated various one-dimensional coding schemes in populations of 1,000 neurons, and visualized them by random projection into 3d space (Fig. 4). The stimulus was parameterized by a single circular variable, like the direction of a grating. A low-dimensional code with two nonzero eigenvalues produced a circular neural manifold (Fig. 4a). An uncorrelated, high-dimensional code where each neuron responded to a different stimulus produced 1,000 equal variances, consistent with the efficient coding hypothesis (Fig. 4b). However this code did not respect distances: responses to stimuli separated by just a few degrees differed as much as responses to diametrically opposite stimuli, and the neural manifold appeared as a spiky, discontinuous ball. Power-law codes (Supplementary Discussion §2.7, Example 2) show a scale-free geometry, whose smoothness depends on the exponent α (Fig. 4c-e). A power-law code with $\alpha = 2$ (Fig. 4c) was a non-differentiable fractal because the many dimensions encoding fine stimulus details (blue) together outweighed the few dimensions encoding large-scale stimulus differences (red). At the critical exponent of $\alpha = 3$ (which equals $1 + 2/d$ since $d = 1$), the neural manifold was on the border of differentiability; the code represents fine differences between stimuli while still preserving large-scale stimulus features (Fig. 4d). A higher exponent led to a smoother neural manifold (Fig. 4e).

What are the computational consequences of these different coding geometries, and what advantage might the brain gain from a power-law code with a close-to-critical exponent? The efficient coding hypothesis suggests information is optimally encoded when responses to different stimuli are as different as possible. However, such codes carry a cost in terms of generalization and robustness: if the neural responses to any pair of stimuli were orthogonal, then stimuli that differ only in tiny details would have completely different representations (Supplementary Discussion §2.1). Similar behavior can be seen in deep neural network architectures that misclassify "adversarial images" differing only very slightly from training examples [34, 35]. We suggest that a power-law code just above the critical exponent represents a balance between efficient, high-dimensional codes, and smooth codes enabling generalization.

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Supplementary Information

Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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Author Contributions

Conceptualization, C.S., M.P., N.S, M.C. and K.D.H.; Methodology, C.S., M.P., N.S. and K.D.H; Software, C.S. and M.P.; Investigation, C.S., M.P., N.S. and K.D.H; Writing, C.S., M.P., N.S., M.C. and K.D.H; Resources, M.C. and K.D.H. Funding acquisition, M.C. and K.D.H.

Author Information

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The authors declare no competing interests.

Materials & Correspondence

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All of the processed deconvolved calcium traces are available on figshare[36] (figshare.com/articles/Recordings_of_ten_thousand_neurons_in_visual_cortex_in_response_to_2_800_natural_images/6845348), together with the image stimuli. The code is available on github (github.com/MouseLand/stringer-pachitariu-et-al-2018b).

Methods

All experimental procedures were conducted according to the UK Animals Scientific Procedures Act (1986). Experiments were performed at University College London under personal and project licenses released by the Home Office following appropriate ethics review.

Animals and surgery

We used mice bred to express GCaMP6s in excitatory neurons in our recordings: 13 recordings from TetO-GCaMP6s x Emx1-IRES-Cre mice (available as JAX 024742 and JAX 005628); 3 recordings from a Camk2a-tTA, Ai94 GCaMP6s 2tg x Emx1-IRES-Cre mouse (available as JAX 024115 and JAX 005628); and 2 recordings from a Camk2a-tTA, Ai94 GCaMP6s 2tg x Rasgrf-Cre mouse (available as JAX 024115 and JAX 022864). We also used mice bred to express tdTomato in inhibitory neurons (GAD-IRES-Cre x CAG-tdTomato, available as JAX 010802 and JAX 007909) in 14 recordings. In this case, GCaMP6s was expressed virally, and excitatory neurons were identified by lack of tdTomato expression. These mice were male and female, and ranged from age 2 to 8 months. We recorded from sufficiently many mice to draw scientific conclusions (8 mice in total). There was no randomization or blinding done in the study.

Surgical methods were similar to those described elsewhere [19, 37]. Briefly, surgeries were performed in adult mice (P35–P125) under isoflurane anesthesia (5% for induction, 0.5-1% during the surgery) in a stereotaxic frame. Before surgery, Rimadyl was administered as a systemic analgesic and lidocaine was administered locally at the surgery site. During the surgery we implanted a head-plate for later head-fixation, and made a craniotomy of 3-4 mm in diameter with a cranial window implant for optical access. In Gad-Cre x tdTomato transgenics, we targeted virus injections (AAV2/1-hSyn-GCaMP6s, University of Pennsylvania Vector Core, 50-200 nl, 1-3 x 1012 GC/ml) to monocular V1

(2.1-3.3 mm laterally and 3.5-4.0mm posteriorly from Bregma), using a beveled micropipette and a Nanoject II injector (Drummond Scientific Company, Broomall, PA 1) attached to a stereotaxic micromanipulator. To obtain large fields of view for imaging, we typically performed 4-8 injections at nearby locations, at multiple depths ($\sim$500 μm and $\sim$200 μm). Rimadyl was then used as a post-operative analgesic for three days, delivered to the mice via their drinking water.

Data acquisition

We used a 2-photon microscope (Bergamo II multiphoton imaging microscope, Thorlabs, Germany) to record neural activity, and ScanImage [38] for data acquisition, obtaining 10622 $\pm$ 1690 (standard deviation) neurons in the recordings. The recordings were performed using multi-plane acquisition controlled by a resonance scanner, with planes spaced 30-35 μm apart in depth. Ten or twelve planes were acquired sequentially, scanning the entire stack repeatedly at 3 or 2.5 Hz. Because plane scanning was not synchronized to stimulus presentation, we aligned the stimulus onsets to each of the planes separately, and computed stimulus responses from the first two frames acquired after stimulus onset for each plane.

The mice were free to run on an air-floating ball and were surrounded by three computer monitors arranged at 90^o angles to the left, front and right of the animal, so that the animal's head was approximately in the geometric center of the setup. Data from running and non-running periods was analyzed together.

For each mouse, recordings were made over multiple days, always returning to the same field of view (in one mouse, two fields of view were used). For each mouse, a field of view was selected on the first recording day such that 10,000 neurons could be observed, with clear calcium transients and a retinotopic location (identified by neuropil fluorescence) localized on the monitors. If more than one potential field of view satisfied these criteria, we chose either a horizontally and vertically central retinotopic location, or a lateral retinotopic location, at 90^o from the center, but still centered vertically. The retinotopic location of the field of view (central or lateral) was unrelated to variance spectra. We also did not observe significant differences between recordings obtained from different modes of GCaMP expression (transgenic vs viral injection). Thus, we pooled data over all conditions.

Visual stimuli

During two-photon recordings, all stimuli other than sparse noise stimuli were presented for 0.5 sec, alternating with a gray-screen inter-stimulus interval lasting a random time between 0.3 and 1.1 s. During electrophysiological recordings, all stimuli were presented for 400ms, with a uniformly-distributed inter-stimulus interval of 300-700ms.

Image stimuli were selected from the ImageNet database [20], from ethologically-relevant categories: "birds", "cat", "flowers", "hamster", "holes", "insects", "mice", "mushrooms", "nests", "pellets", "snakes", "wildcat". Images were chosen manually to ensure that less than 50% of the image was a uniform background, and to contain a mixture of low and high spatial frequencies. The images were uniformly contrast normalized. This was achieved by subtracting the local mean brightness and dividing by the local mean contrast (standard deviation of pixel values); the local mean and standard deviation were both computed using a Gaussian filter of standard deviation 30°. Each presented stimulus consisted of a different normalized image from ImageNet (2800 different images) replicated across all 3 screens, but at a different rotation on each screen (Fig. 1c).

For the main two-photon recordings, these 2800 stimuli were presented twice, in the same order each time. In the electrophysiological recordings, 700 of these same stimuli were presented twice in the same order each time. Additionally, in a subset of imaged mice (4 out of 6), we presented a smaller set of 32 images, presented in a randomized order 90-114 times, to enable more accurate estimation of trial-averaged responses.

We also presented partially spatially whitened versions of the 2800 natural images. To compute spatially whitened images, we first computed the two-dimensional Fourier spectrum for each image, and averaged the spectra across images. We then whitened each image in the frequency domain by dividing its Fourier transform by the averaged

Fourier spectrum across all images. The rescaled Fourier transform of the image was transformed back into the pixel domain by computing its inverse two-dimensional Fourier transform and retaining the real part. Each image was then intensity-scaled to have the same mean and standard deviation pixel values as the original.

The eight- and four-dimensional stimuli were constructed using a reduced-rank regression model. We first used reduced rank regression to predict the neuronal population responses R from the natural images I (N_{pixels} by N_{stimuli}) via a d-dimensional bottleneck :

$$R = A^T BI$$

where A is a matrix of size d by N_{neurons} and B is a matrix of size d by N_{pixels}. The dimensionality d was either eight or four depending on the set of stimuli being constructed. The columns of B represent the image dimensions which linearly explain the most variance in the neural population responses. The stimuli were the original 2800 natural images projected onto the reduced-rank subspace B: $I_{\text{low-D}} = B^\top BI$.

In addition to natural image stimuli, we also presented drifting gratings and sparse noise. Drifting gratings of 32 directions, spaced evenly at directions evenly spaced at 11°, were presented 70-128 times each, lasting 0.5 sec each, and with a gray-screen inter-stimulus interval between 0.3 and 1.1 s. They were full-field stimuli (all three monitors) and their spatial frequency was 0.05 cycles per degree and their temporal frequency was 2Hz.

Sparse noise stimuli consisted of white or black squares on a gray background. Squares were of size 5°, and changed intensity every 200 ms. On each frame, the intensity of each square was chosen independently, as white with 2.5% probability, black with 2.5% probability, and gray with 95% probability. The sparse noise movie contained 6000 frames, lasting 20 minutes, and the same movie was played twice to allow cross-validated analysis.

Spontaneous activity was recorded for 30 minutes with all monitors showing a gray or black background, in all but six of 32 image set recordings. In three recordings of 32-natural image responses and three recordings of drifting grating responses, we interspersed the spontaneous activity, recording 30 seconds of spontaneous gray screen activity in between each set of 32 stimuli. In all recordings but these six, there were also occasional blank stimuli (1 out of every 20 stimuli in the 2800 natural image stimuli). The activity during these non-stimulus periods was used to project out spontaneous dimensions from the neuronal population responses (see below).

Calcium imaging processing

Calcium movie data was processed using the Suite2p toolbox [19, 37], available at www.github.com/cortex-lab/Suite2P.

Briefly, the Suite2p pipeline consists of registration, cell detection, ROI classification, neuropil correction, and spike deconvolution. Movie frames are registered using 2D translation estimated by regularized phase correlation, subpixel interpolation and kriging. To detect regions of interest (ROIs; corresponding to cells), Suite2p clusters correlated pixels, using a low-dimensional decomposition of the data to accelerate processing. The number of ROIs is determined automatically via a threshold on pixel correlations. Finally, ROIs were classified as somatic or non-somatic using a classifier trained on a set of human-curated ROIs. The classifier reached 95% agreement with training data, thus allowing us to skip manual curation for most recordings. For neuropil correction, we used the approach of Ref. [39], subtracting from each ROI signal the surrounding neuropil signal scaled by a factor of 0.7; all pixels attributed to an ROI (somatic or not) were excluded from the neuropil trace. After neuropil subtraction, we further subtracted a running baseline of the calcium traces with a sliding window of 60 seconds to remove long timescale additive shifts in the signals. Fluorescence transients were estimated using non-negative spike deconvolution [40] with a fixed timescale of calcium indicator decay of 2 seconds, a method which we found to out-perform others on ground truth data [41]. Finally, each cell's deconvolved trace was z-scored with respect to the mean and standard deviation of that cell's trace during a 30-minute period of gray-screen spontaneous activity before or after the image presentations.

All of the processed deconvolved calcium traces are available on figshare[36] (https://figshare.com/articles/Recordings_of_ten_thousand_neurons_in_visual_cortex_in_response_to_2_800_natural_images/6845348), together with the image stimuli.

Data acquisition and processing (electrophysiology)

Neuropixels electrode arrays [42] were used to record extracellularly from neurons in six mice. The mice were between 8 and 24 weeks old at the time of recording, and were of either gender. The genotypes of the mice were Slc17a7-Cre;Ai95, Snap25-GCaMP6s, TetO-GCaMP6s;CaMKIIa-tTA, Ai32;Pvalb-Cre (two mice), or Emx1-Cre;CaMKIIa-tTA;Ai94. In some cases, other electrophysiological recordings had been made from other locations in the days preceding the recordings reported here. In all cases, a brief (<1 hour) surgery to implant a steel headplate and 3D-printed plastic recording chamber (12mm diameter) was first performed. Following recovery, mice were acclimated to head-fixation in the recording setup. During head-fixation, mice were seated on a plastic apparatus with forepaws on a rotating rubber wheel (five mice) or were on a styrofoam treadmill and able to run (one mouse). Three 20x16cm TFT-LCD screens (LG LP097QX1) were positioned around the mouse at right angles at a distance of 10cm, covering a total of 270x78 degrees visual angle. On the day of recording, mice were again briefly anesthetized with isoflurane while eight small craniotomies were made with a dental drill. After several hours of recovery, mice were head-fixed in the setup. Probes had a silver wire soldered onto the reference pad and shorted to ground; these reference wires were connected to a Ag/AgCl wire positioned on the skull. The craniotomies as well as the wire were covered with saline-based agar. The agar was covered with silicone oil to prevent drying. Probes were each mounted on a rod held by an electronically position-able micromanipulator (uMP-4, Sensapex Inc.) and were then advanced through the agar and through the dura. Once electrodes punctured dura, they were advanced slowly (10μm/sec) to their final depth (4 or 5 mm deep). Electrodes were allowed to settle for approximately 15 minutes before starting recording. Recordings were made in external reference mode with LFP gain=250 and AP gain=500, using SpikeGLX software. Data were preprocessed by re-referencing to the common median across all channels. 6 recordings were performed in 6 different mice, with a total of 14 probes in visual cortex across all experiments.

We spike sorted the data using a modification of Kilosort [43] that tracks drifting clusters, called Kilosort2 [37, 44], available at www.github.com/MouseLand/Kilosort2. Without the modifications, the original Kilosort and similar algorithms can split clusters according to drift of the electrode. Kilosort2 in comparison tracks neurons across drift levels and for longer periods of time ($\sim$ 1 hour in our case).

Removal of ongoing activity dimensions

As shown previously [37], approximately half the variance of visual cortical population activity is unrelated to visual stimuli, but represents behavior-related fluctuations. This ongoing activity continues uninterrupted during stimulus presentations, and overlaps with stimulus responses only along a single dimension. Because the present study is purely focused on sensory responses, we projected out the dimensions corresponding to ongoing activity prior to further analysis. The top 32 dimensions of ongoing activity were found by performing principal component analysis on the z-scored ongoing neural activity recorded during a 30-minute period of gray screen stimuli before or after the image presentations. To remove these dimensions from stimulus responses, the stimulus-driven activity was also first z-scored (using the mean and variance of each neuron computed from ongoing activity), then the projection onto the 32 top spontaneous dimensions was subtracted (Extended Data Fig. 4).

In the electrophysiological recordings, we considered stimulus responses in a window of 50 ms or 500 ms following stimulus onset. Therefore, we computed the ongoing activity using these two different bin sizes (50 ms or 500 ms). Then we z-scored the stimulus responses by this ongoing activity. Next we computed the top 10 principal components of the ongoing activity (in both bin sizes) and then subtracted the projection of the stimulus responses onto these dimensions.

Receptive field estimation

We estimated the receptive fields of the neurons, either using a reduced-rank regression model or using a simple/complex Gabor model. In both cases, the model was fit to the mean response of each neuron to half of the

2800 images (I_{train}) over the two repeats. The performance of the model was tested on the mean response of each neuron to the other half of the 2800 images (I_{test}).

Reduced-rank receptive field estimation

To estimate a linear receptive field for each neuron, we used reduced rank regression [45], a self-regularizing method which allowed us to fit all neurons' responses to a single repeat of all 2800 image stimuli. Reduced rank regression predicts high-dimensional outputs from high-dimensional inputs through a linear low-dimensional hidden "bottleneck" representation. We used a 25-dimensional hidden representation to predict each neuron's activity from the image pixel vectors, taking the resulting regressor matrices as the linear receptive fields. These receptive fields explained 11.4±0.7% (SE, n=7 recordings) of the stimulus-related variance on the test set. These were z-scored prior to display in Fig. 1h and Extended Data Fig. 3a.

Model-based receptive field estimation

To fit classical simple/complex receptive fields to each cell, we simulated the responses of a convolutional grid of Gabor filters to the natural images, and fit each neuron with the the filter response most correlated to its response.

The Gabor cell filters $G(\mathbf{x})$ were parametrized by a spatial frequency f, orientation θ, phase ψ, size α and eccentricity β. Defining $\mathbf{u}$ and $\mathbf{v}$ to be unit vectors pointing parallel and perpendicular to the orientation θ:

$$G(\mathbf{x}) = cos(2\pi f \mathbf{x} \cdot \mathbf{u} + \psi) e^{-((\mathbf{x}\cdot\mathbf{u})^2 + \beta(\mathbf{x}\cdot\mathbf{v})^2)/2\alpha^2}$$

We constructed 12288 Gabor filters, with centers spanning a 9 by 7 grid spaced at 5 pixels, and with parameters f, θ, ϕ, α, and β ranging from $(0.01, 0, 0, 3, 1)$ to $(0.13, 157, 315, 12, 2.5)$ with $(7, 8, 8, 4, 4)$ points sampled of each parameter respectively. The parameters were equally spaced along the grid (e.g. f was sampled at 0.01, 0.03, 0.05, 0.07, 0.09, 0.11, 0.13).

Simple cell responses were simulated by passing the dot product of the image with the filter through a rectifier function $r(x) = max(0, x)$. Complex cell responses were simulated as the root-mean-square response of each unrectified simple cell filter and the same filter with phase ψ shifted by 90°. A neuron's activity was predicted as a linear combination of a simple cell and its complex cell counterpart, with weights estimated by linear regression. Each neuron was assigned to the filter which best predicted its responses to the training images (Extended Data Fig. 3b-h). This simple/complex Gabor model explained 18.4±0.1% of the stimulus-related variance on the test set.

We also evaluated a model of Gabor receptive fields including divisive normalization [46]. To do so, the response of each of modeled simple or complex cell filter was divided by the summed, normalized responses of all the other simple and complex cells at this retinotopic location. The experimentally measured response of each neuron was then predicted as a linear combination of simple and complex responses to the best-fitting Gabor, with weights estimated by linear regression. 45.4% ± 1.0% (mean ± SE) of cells were better fit by the divisive normalization model. However, while divisive normalization changed the optimal parameters fit to many cells (Extended Data Fig. 3i-n), the resulting eigenspectra were indistinguishable from a model with no normalization (Extended Data Fig. 3o-u).

Sparseness estimation

To estimate the sparseness of single-cell responses to the image stimuli, we counted how many neurons were driven more than two standard deviations above their baseline rate by any given stimulus. This was estimated using 4 experiments in which 32 natural images were repeated >90 times. We computed each neuron's tuning curve by averaging over all repeats. The standard deviation of the tuning curve is computed for each neuron across stimuli. Baseline rate was defined as the mean firing rate during all spontaneous activity periods, without visual stimuli. A neuron was judged as responsive to a given stimulus if its response was more than two times this standard deviation plus its baseline firing rate.

Decoding accuracy from 2,800 stimuli

To decode the stimulus identity from the neural responses (Fig. 1g), we built a simple nearest neighbor decoder based on correlation. The first stimulus presentation was used as training set while the second presentation was used as test set. We correlated the population responses for a individual stimulus in the test set with the population responses from all stimuli in the training set. The stimulus with the maximum correlation was then assigned as our prediction. We defined the decoding accuracy as the fraction of correctly labelled stimuli.

Signal-to-noise ratio and explained variance

To compute the tuning-related signal-to-noise ratio (SNR; Fig. 1f), we first estimated the signal variance of each neuron $\hat{V}_{\text{sig}}$ as the covariance of its response to all stimuli across two repeats (for neuron c, $\hat{V}_{\text{sig}} = \text{Cov}_s[f_1(c,s), f_2(c,s)]$, see Supplementary Discussion §1). The noise variance $\hat{V}_{\text{noise}} = V_{\text{tot}} - \hat{V}_{\text{sig}}$ was defined as the difference between the within-repeat variance (reflecting both signal and noise) and this signal variance estimate, and the SNR as their ratio. The SNR estimate is positive when a neuron has responses to stimuli above its noise baseline; note that as $\hat{V}_{\text{sig}}$ is an unbiased estimate, it can take negative values when he true signal variance is zero.

To compute the percentage of explained variance for each neuron (Extended Data Fig. 1c), we divided the estimated signal variance by the total variance across trials (averaged across the repeats):

$$EV = \frac{\hat{V}_{\text{sig}}}{\frac{1}{N_r}\sum_r \text{Var}_t[f_{t,r}]}.$$

Note that this formula is similar to the Pearson correlation of a neuron's responses between two repeats. In Pearson correlation, the numerator is the same, equal to the covariance between repeats, but the denominator is the geometric rather than arithmetic mean of the variances of the two repeats.

cvPCA method

The cvPCA method is fully described in Supplementary Discussion §1, characterized mathematically in §1.1 and §3.6, and analyzed in simulation in Extended Data Fig. 5. In brief, the difference between this approach and standard PCA (e.g. Refs. [47, 48]) is that it compares the activity on training and test repeats to obtain an estimate of the stimulus-related ("signal") variance, discounting variance from trial-to-trial variability ("noise").

Denote the response of neuron c to repeat r of stimulus s by $f_r(c,s)$, define the signal as the expected response, which will be equal for both repeats: $\phi(c,s) = \mathbb{E}\left[f_r(c,s)\right]$, and the noise on repeat r to be the residual after the expected response is subtracted: $\nu_r(c,s) = f_r(c,s) - \phi(c,s)$. By definition, the noise has zero expectation: $\mathbb{E}_{\nu_r}\left[\nu_r(c,s)|c,s\right] = 0$ for all r, c, and s. Let $\hat{\mathbf{u}}_n$ denote the n^{th} principal component eigenvector, computed from repeat 1.

If we estimated the variance of the projection of activity onto $\hat{\mathbf{u}}_n$ using a single repeat, it would contain a contribution from both the signal and the noise. However, because stimulus-independent variability is by definition uncorrelated between repeats of the same stimulus, we can obtain an unbiased estimate of the signal variance, from the covariance across these independent repeats:

$$\mathbb{E}_{\nu_1,\nu_2}\left[\frac{1}{N_s}\sum_{i=1}^{N_s}(\mathbf{f}_1(s_i)\cdot\hat{\mathbf{u}}_n)(\mathbf{f}_2(s_i)\cdot\hat{\mathbf{u}}_n)\right]$$
$$=\frac{1}{N_s}\sum_{i=1}^{N_s}(\boldsymbol{\phi}(s_i)\cdot\hat{\mathbf{u}}_n)^2$$

Thus, if $\mathbf{u}_n$ is an eigenvector of the population signal variance, the cvPCA method will produce an unbiased estimate of the signal principal component variances. As shown in Supplementary Discussion 1.1, this will occur if response variability comprises a mixture of multiplicative response gain changes, correlated additive variability orthogonal to the stimulus dimensions, and uncorrelated noise. Although additive variability in the signal space could in principle downwardly bias the estimated signal variance, other work confirms that under conditions similar to those analyzed here there is little additive variability in the signal space [37]; furthermore, simulations confirm that the amount of such variability present in our recordings does not substantially bias the estimation of signal eigenspectra with cvPCA (Extended Data Fig. 5).

We ran cvPCA 10 times on each dataset, on each iteration randomly sampling each stimulus' population response from the two repeats without replacement. Thus, $\mathbf{f}_1(s)$ could be the population response from either the first or second repeat, with $\mathbf{f}_2(s)$ being the response from the other. The displayed eigenspectra are averages over the 10 different runs.

Simulations

To verify that cvPCA method was able to accurately estimate signal eigenspectra in the presence of noise, we analyzed simulated data where the true eigenspectrum was known by construction, and stimulus responses were corrupted by noise. Mathematical analyses (Supplementary Discussion 1.1 and 3.6) showed that noise consisting of multiplicative gain modulation, additive noise orthogonal to signal dimensions, or independent additive noise should not bias the expected eigenspectrum estimate, but that correlated additive noise in the stimulus dimensions could potentially lead the eigenspectrum to be underestimated. We therefore first concentrated on this possibility.

To create the test data, we first simulated noise-free sensory responses whose eigenspectrum followed an exact power law, with three possible exponents: $\alpha = 0.5$, 1.0, or 1.5. To simulate the responses of $N_c = 10,000$ neurons to $N_s = 2,800$ stimuli with this exact eigenspectrum, we first constructed a set of random orthogonal eigenvectors by performing singular value decomposition on a $N_c \times N_s$ matrix A of independent standard Gaussian variates: $A = USV^\top$. We created a diagonal matrix D_α, whose n^{th} diagonal entry was $n^{-\alpha/2}$, and created the $N_c \times N_s$ matrix of simulated noise-free responses as $\boldsymbol{\phi} = UDV^\top$.

Additive noise

To simulate correlated additive noise in the stimulus space (Extended Data Fig. 5c), we constructed noise whose eigenspectrum matched that observed experimentally. To find the empirical noise eigenspectrum, we first estimated the total variance of the n^{th} principal component as

$$\hat{\Lambda}_n = \frac{1}{2}\left[\frac{1}{N_s}\sum_{i=1}^{N_s}(\mathbf{f}_1(s_i)\cdot\mathbf{u}_n)^2 + \frac{1}{N_s}\sum_{i=1}^{N_s}(\mathbf{f}_2(s_i)\cdot\mathbf{u}_n)^2\right]$$

and estimated the signal variance using cvPCA as

$$\hat{\lambda}_n = \frac{1}{N_s}\sum_{i=1}^{N_s}(\mathbf{f}_1(s_i)\cdot\mathbf{u}_n)(\mathbf{f}_2(s_i)\cdot\mathbf{u}_n).$$

The estimated noise spectrum was the difference between total variance and noise variance: $\hat{\delta}_n = \hat{\Lambda}_n - \hat{\lambda}_n$. This spectrum reflects the summed magnitude of both correlated and uncorrrelated noise in the signal dimensions, and is shown in Extended Data Fig. 5b. Responses corrupted by additive noise were simulated as $\boldsymbol{\phi} + b_\alpha U \Delta V^\top$, where Δ is a diagonal matrix with entries δ_n, and the scale factor b_α ensured that, as in the data, the simulation showed a total of 14% reliable variance. The scale factors were found by search to be 2.62, 2.52, and 2.41 for the signal eigenspectrum exponents $\alpha = 0.5$, 1.0, and 1.5, respectively.

Multiplicative noise

To simulate multiplicative noise (Extended Data Fig. 5d), responses were multiplied by an amplitude factor that was constant across neurons, but was drawn independently for each stimulus and repeat. To simulate an appropriately skewed distribution of gains, the scale factor was distributed as 0.5 plus an exponential random variate with mean parameter c_α. The values of c_α were found by search as those matching the observed 14% reliable variance, yielding 1.55, 1.52, and 1.40 for the signal eigenspectrum exponents $\alpha = 0.5$, 1.0, and 1.5, respectively.

To simulate a combination of additive and multiplicative noise (Extended Data Fig. 5e), responses were modulated by the additive mechanism described above and then modulated multiplicatively. The gain factors were $b_\alpha = 0.55, 0.53, 0.51$ and $c_\alpha = 0.65, 0.64, 0.59$ for $\alpha = 0.5, 1.0, 1.5$ respectively.

Two-photon noise

To investigate whether our two-photon deconvolution method could be biasing the estimated eigenspectrum, we simulated the effect of passing noise through this algorithm (Extended Data Fig. 5f).

To do so, we extended the simulations above to apply in the time domain. When simulating the additive noise, we allowed it to vary across all simulated 2-photon imaging frames (replacing the matrix A used to compute the eigenvectors U and V by a 10000×8400 matrix providing 3 simulated frames per stimulus presentation). The gain modulation factor was assumed equal for all three frames corresponding to a single stimulus. The magnitudes of the additive noise and the gain factor giving 14% signal variance were found by search to be $b_\alpha = 0.50, 0.50, 0.49$ and $c_\alpha = 0.68, 0.67, 0.66$, for $\alpha = 0.5, 1.0, 1.5$ respectively.

To simulate the GCaMP6s response, we convolved these responses with an exponentially decaying kernel with a timescale of 2 seconds (since each timepoint in the data is 0.4 seconds, this corresponds to a decay timescale of 5 timepoints). To simulate shot noise, we added Gaussian white noise with a standard deviation of 0.5. Next we deconvolved these noisy traces using OASIS [40], with a timescale of 5 timepoints and no sparsity constraints. The reduction in signal variance from this procedure was roughly 1%.

For all noise simulations, we estimated the signal eigenspectrum from two repeats using cvPCA. We found that cvPCA, but not ordinary PCA, correctly estimated the ground-true eigenspectrum, for all simulated power-law exponents α (Extended Data Fig. 5g).

Estimation of power-law exponent

We computed the linear fit of the eigenspectrum over the range of 11 to 500 dimensions for all recordings (and model fits) other than the 32 drifting grating recordings. For the 32-grating recordings, due to noise and the length of the spectrum, we computed the power-law exponent from 5 to 30. The linear fit was performed in log-log space: the range of log(11) to log(500) was regressed onto the log of the eigenspectrum.

Sorting neurons and stimuli by correlations

In Extended Data Fig. 6, neurons and stimuli were sorted so they were close to other neurons and stimuli which which they were correlated.

To do this, we first z-scored the binned activity of each neuron and computed principal components of its averaged activity across repeats. Each panel shows this for different PC projections of the data: 1,2, 3-10, 11-40, 41-200, and 201-1000. Stimuli were re-ordered so that each stimulus's pattern of evoked population activity was most similar to the average of its neighbors. The stimulus order was initialized by sorting stimuli according to their weights on the top principal component of activity, then dividing them into 30 clusters of equal size along this ordering. For 50 iterations, we computed the mean activity of each cluster and smoothed this activity across clusters with a Gaussian whose width annealed from 6 clusters to 1 over the first 25 iterations. Each stimulus was then reassigned to the cluster it was most correlated with. On the final pass, we upsampled the stimuli's correlations with each cluster by a factor of 100 via kriging interpolation (smoothing constant of 1 cluster), resulting in a continuous assignment of stimuli along the 1D axis of clustering algorithm. After sorting across stimuli, we smoothed across them to reduce noise, recomputed the principal components on the activity smoothed across stimuli, and repeated the procedure to sort neurons. The algorithm is available in Python and MATLAB at www.github.com/MouseLand/RasterMap. These plots were made using the MATLAB version of the code.

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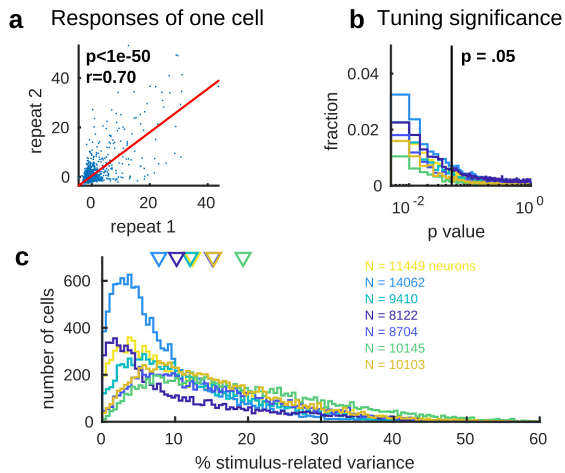
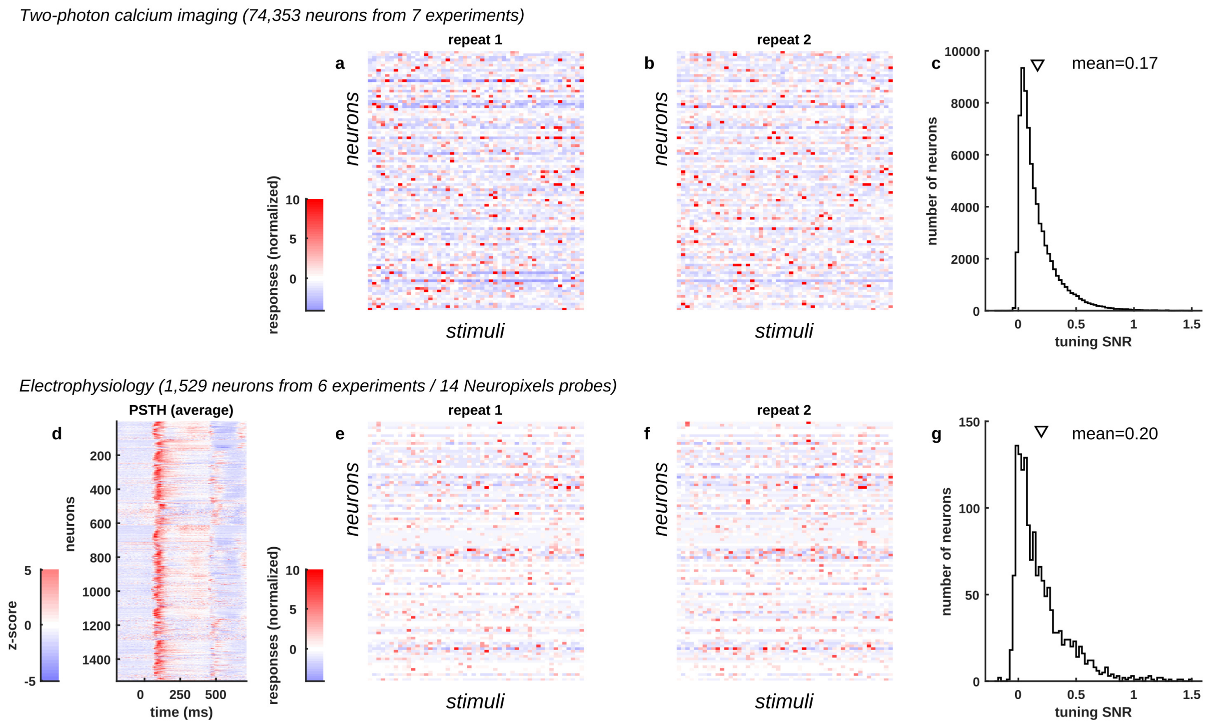
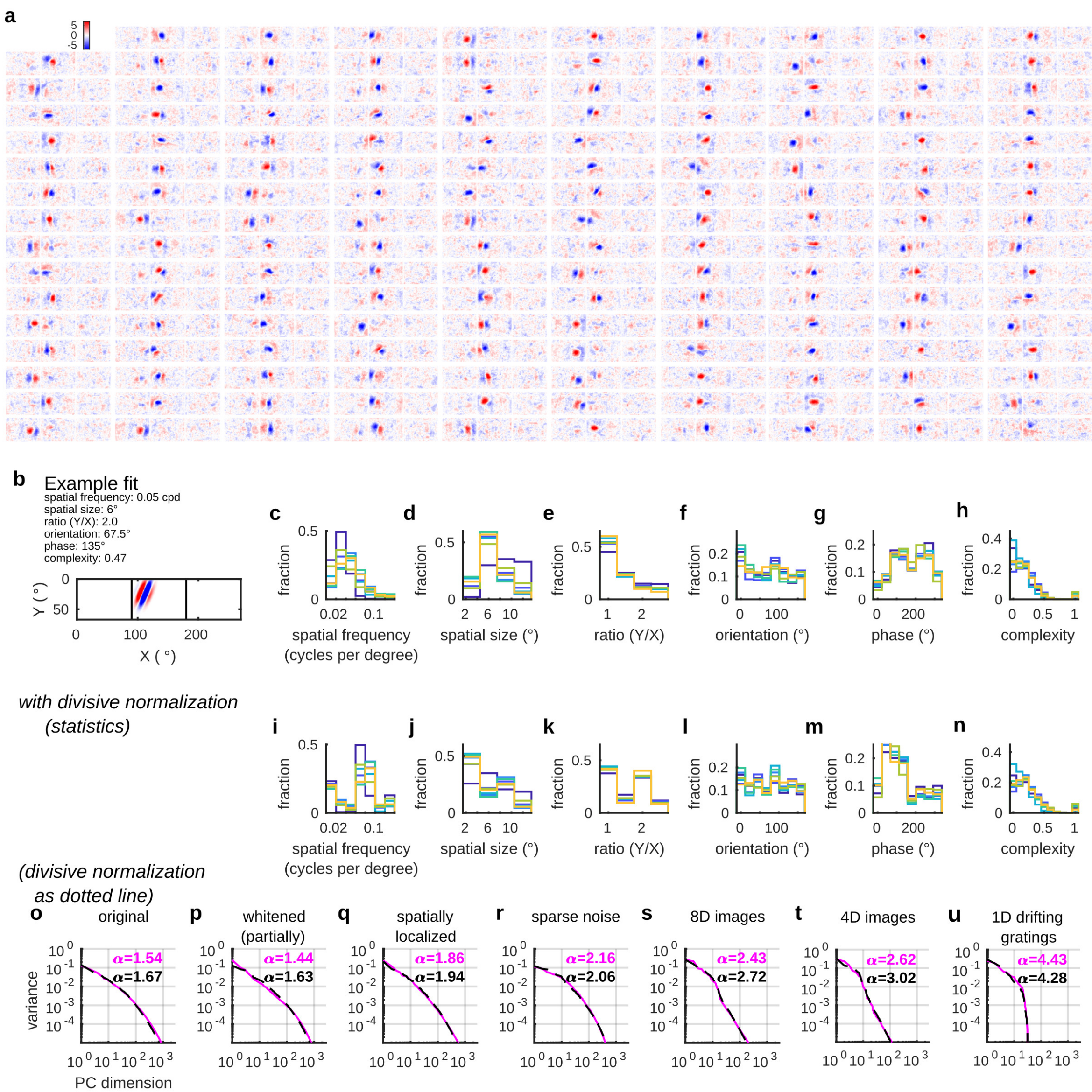


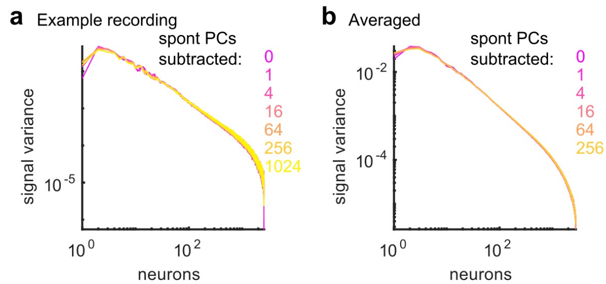
Extended Data Fig. 1 | Reliability of single neuron responses. **a**, A single neuron's response to the first repeat of 2800 stimuli plotted against its responses to the second repeat of the same stimuli. **b**, Histograms of p-values for Pearson correlation of responses on the two repeats. Each colored histogram represents a different recording. 81.4 $\pm$ 5.1% (SE, n=7 recordings) of cells were significant at $p<0.05$. **c**, Histogram of the single neuron percentage of stimulus-related variance across the population. Each colored histogram represents a different recording; arrowheads (top) represent the mean for each experiment.



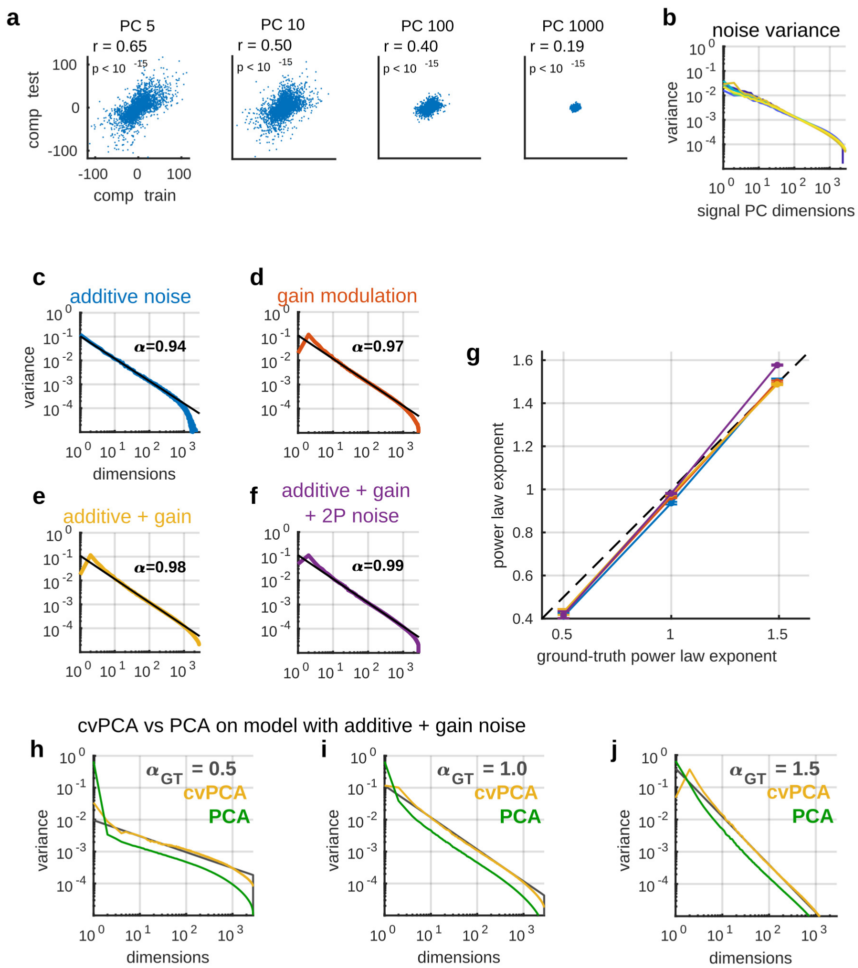
Extended Data Fig. 2 | Comparison with electrophysiology. a,b, Single trial responses of 100 neurons to two repeats of 50 stimuli, recorded by two-photon calcium imaging. **c,** Distribution of tuning SNR for 74,353 neurons recorded by two-photon calcium imaging. **d,** Average peri-stimulus time histogram of spikes recorded electrophysiologically in a separate set of experiments. The images shown were a random subset of 700 images out of the total 2,800. The PSTH reflects the average over all stimuli. The responses are z-scored across time for each neuron. **efg** Same as (abc) for the electrophysiologically recorded neurons.



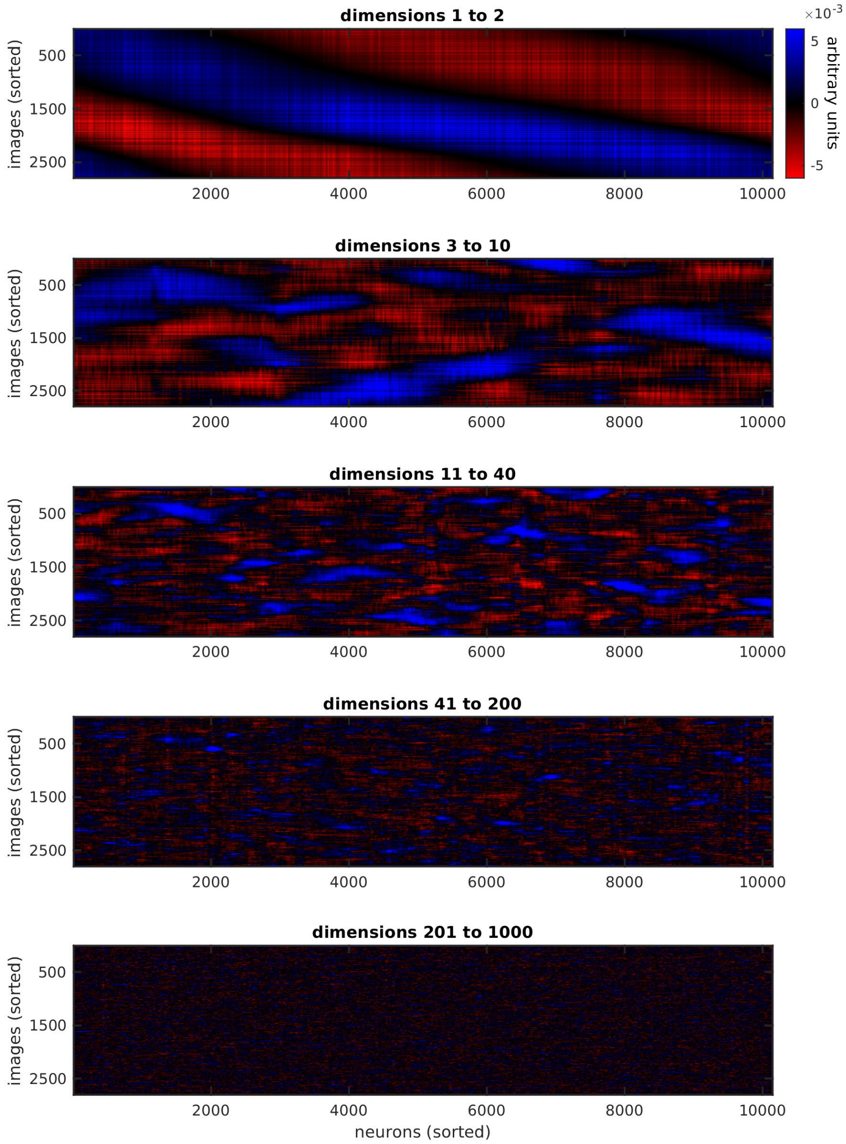
Extended Data Fig. 3 | Single neuron receptive fields estimated using reduced-rank regression and Gabor models. a, 159 randomly chosen neurons' receptive fields estimated using reduced-rank regression. The receptive field map is z-scored for each neuron. **b**, An example Gabor fit to a single cell. **c-h**, Histograms showing the distribution of model parameters across cells. Each color represents cells from one recording. **i-n**, Histograms showing the distribution of model parameters across cells when model also has divisive normalization. **o-u**, Eigenspectra of Gabor population model responses to the different stimulus sets. The unnormalized Gabors are shown in magenta, and the model with divisive normalization in black.



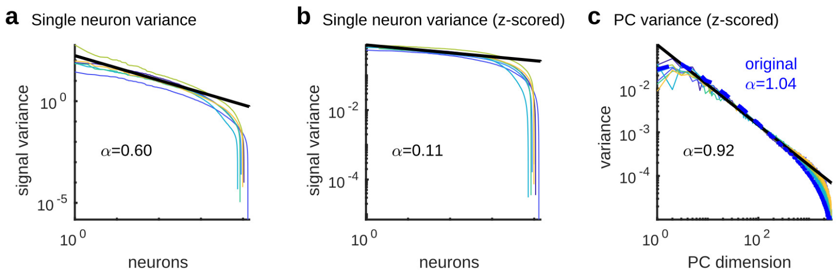
Extended Data Fig. 4 | Stimulus-independent activity does not affect the measured eigenspectrum. a, To measure the effects of correlated noise variability on eigenspectra estimated by cvPCA, we examined the effect of projecting out different numbers of noise dimensions (estimated during periods of spontaneous gray-screen) from the responses in an example experiment. **b**, Same analysis, averaged over all recordings. The presence of these noise dimensions made little difference to the estimated signal eigenspectrum other than to slightly reduce estimated eigenvalues in the highest and lowest dimensions. For the main analyses, 32 spontaneous dimensions were subtracted.



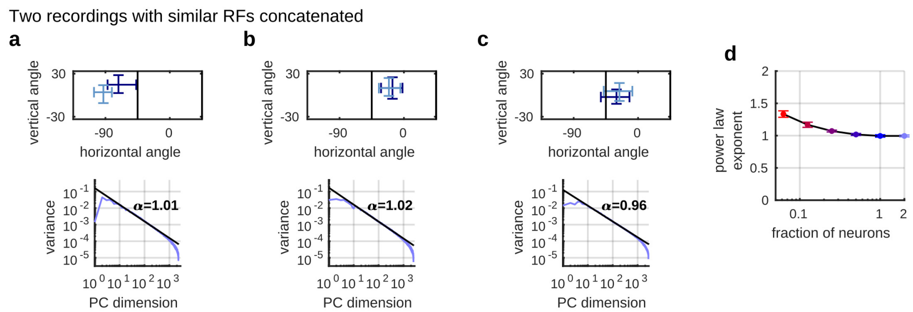
Extended Data Fig. 5 | Validating the eigenspectrum estimation method using simulations with the true noise distribution. **a**, Scatterplots illustrating the noise levels of each estimated PC. Each plot shows population activity projected onto the specified principal component, for the 1st repeat (x-axis) and 2nd repeat (y-axis). Each point represents responses a a single stimulus. **b**, Estimated level of noise variance in successive signal dimensions. Noise variance was estimated by subtracting the cvPCA estimate of signal variance from the total variance (see Methods). **c**, Recovery of ground-truth eigenspectrum in simulated data. We simulated responses of 10,000 neurons to 2,800 stimuli with a power spectrum decay of exactly $\alpha = 1$, and added noise in the stimulus space, generated with the spectrum in (b) scaled to produce the same signal-to-noise ratio as in the original neural data. The ground-truth eigenspectrum (black) is estimated accurately by the cvPCA method (blue). **d**, Same analysis with multiplicative noise, in which the responses of all neurons on each trial scaled by a common random factor. The distribution of this factor was again scaled to recover the original signal-to-noise ratio. **e**, Same analysis with a combination of multiplicative and additive noise. **f**, Same analysis, also including simulation of neural and 2-photon shot noise prior to running GCaMP deconvolution algorithm. **g**, 10 instantiations of the simulation were performed with ground-truth exponents of 0.5, 1.0, and 1.5. Error bars represent standard deviations of the power law exponents estimated for each of the 10 simulations. Dashed black line: ground-truth value. **h-j**, Comparison of cvPCA (yellow) and traditional PCA (green) algorithms in the presence of the additive+multiplicative noise combination. While cvPCA recovered the ground truth eigenspectrum (black) exactly, traditional PCA did not, resulting in overestimation of the top eigenvalues and failure to detect the ground-truth power law.



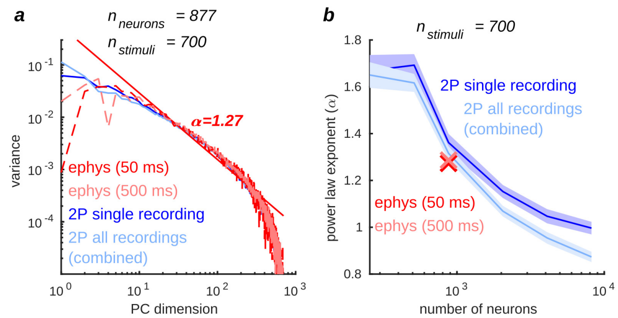
Extended Data Fig. 6 | Successive PC dimensions encode finer stimulus features. Each plot shows the responses of 10,145 neurons to 2,800 natural images, projected onto the specified PCs and then sorted along both axes so correlated neurons and stimuli are close together. We then smoothed the matrix across neurons and stimuli with Gaussian kernels of widths 8 neurons and 2 stimuli respectively. Dimensions 1-2 reveal a coarse, 1-dimensional organization of the neurons and stimuli. Dimensions 3-10 reveal multidimensional structure which involves different neural subpopulations responding to different stimuli. Dimensions 11-40 reveal finer-structured patterns of correlated selectivity among neurons. Dimensions 41-200 and 201-1000 reveal even finer-structured selectivity, which contained less neural variance.



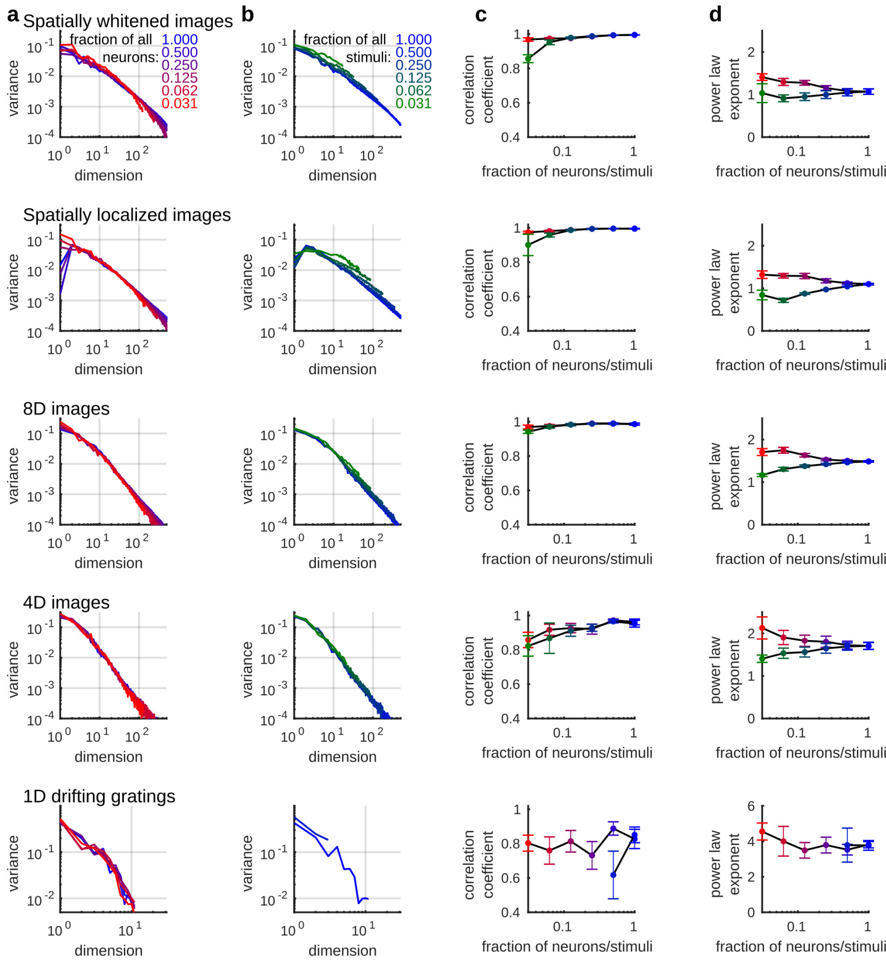
Extended Data Fig. 7 | Power law scaling reflects correlation structure, not single-neuron statistics. **a**, The signal variance of each neuron's responses are sorted in descending order; they approximately follow a power law with a decay exponent of $\alpha = 0.59$. **b**, Same plot after z-scoring the recorded traces to equalize stimulus response sizes between cells; the distribution of single-neuron variance has become nearly flat. **c** PC eigenspectra for z-scored data. Each colored line represents a different recording. Dashed blue shows the average eigenspectrum from the original, non-z-scored responses. The fact that the eigenspectrum power-law is barely affected by equalizing firing rates, while the distribution of single cell signal variance is altered, indicates that the power law arises from correlations between cells rather than from the distribution of firing rates or signal variance across cells.



Extended Data Fig. 8 | Power law eigenspectra in concatenated recordings. **a-c**, To ask whether powerlaw eigenspectra apply to even larger populations, we were able to artificially double the number of recorded neurons by combining three pairs of recordings whose imaging fields of view had similar retinotopic locations. Top: retinotopic locations of receptive fields (95% confidence intervals on that recording's mean RF position), with each recording shown in a different shade of blue. Bottom: eigenspectrum of concatenated recordings in response to the 2800 natural image stimuli; total population sizes 19571, 23472, and 18807 cells respectively. Each column represents one pair of recordings. **d**, Eigenspectrum exponents for random subsets of the combined populations (cf. Fig. 2j). X-axis shows population size relative to single recordings, so merged population is 2. Mean power law exponent at "2x" was $\alpha = 0.99 \pm 0.02$ (mean $\pm$ SE).



Extended Data Fig. 9 | Eigenspectrum of electrophysiologically recorded data. We recorded neural activity electrophysiologically in response to 700 out of the 2800 stimuli, and concatenated the recordings, resulting in a total of 877 neurons recorded across 6 experiments. **a**, With this smaller number of stimuli and neurons, convergence to a power law is not complete, and the exponent cannot be estimated accurately (cf. Fig. 2g-j). We therefore compared the electrophysiology data to a the responses generated by these stimuli in 877 neurons sampled randomly from either a single two-photon imaging experiment (dark blue) or all experiments combined (light blue). Red and pink show electrophysiology eigenspectra with time bins of 50 or 500ms; red line shows best linear fit to estimate exponent. **b**, Blue curves: power law exponents estimated from the responses of different-sized neuronal subpopulations to this set of 700 stimuli (shading: SE over different random subsets of neurons). Red and pink crosses: estimated exponents from electrophysiology data for 50 and 500 ms bin sizes.



Extended Data Fig. 10 | Power law scaling grows more accurate for increasing numbers of neurons and stimuli, for all stimulus ensembles. a, Eigenspectra estimated from a random subset of the recorded neurons (color-coded by fraction of neurons retained. **b**, Eigenspectra estimated from a random subset of stimuli, color-coded by fraction of stimuli retained. **c**, Correlation coefficient of the spectra plotted in **a,b**. **d**, Power law exponent of the spectra plotted in **a,b**. Each row corresponds to a different ensemble of visual stimuli.

Supplementary Discussion

1. Analysis of cvPCA algorithm

Neuronal responses to sensory stimuli can be divided into signal and noise components. If the response of cell c to repeat r of stimulus s is $f_r(c,s)$, we define the "signal" as the expected response, which will be equal for all repeats: $\phi(c,s) = \mathbb{E}\left[f_r(c,s)\right]$, and the "noise" on repeat r to be the residual after the expected response is subtracted: $\nu_r(c,s) = f_r(c,s) - \phi(c,s)$. This terminology is standard but somewhat unfortunate: by this definition, noise represents any difference between the response on a single trial and the trial-average, even though it may reflect deterministic encoding of behavioral or cognitive variables [1, 2, 3]. We assume that the noise is independently and identically distributed across repeats of a single stimulus; this condition can be approximated in practice by separating the presentation of the stimulus repeats by tens of minutes to avoid temporally correlated activity. By definition, the noise has zero expectation: $\mathbb{E}_{\nu_r}\left[\nu_r(c,s)|c,s\right] = 0$ for all r, c, and s. However, we do not assume that the noise has any particular probability distribution, and we allow its distribution and variance to depend on the stimulus.

We cannot measure the noise-free response $\phi(c,s)$ directly. Although it could in principle be estimated by averaging, this requires a very large numbers of repeats. Instead, we focus directly on our quantity of interest, the signal variance (i.e. the variance of the noise-free response across stimuli), for which an unbiased estimate can be obtained from just two repeats. We first describe how cross-validation allows an unbiased estimate of a single cell's signal variance, before moving on to analyze the cvPCA method in full.

Suppose we would like to compute the variance of cell c's response across a set of N_s stimuli, $s_1 \dots s_{N_s}$. To simplify the analysis below, we shift means without loss of generality so that the neuron's mean response is zero: $\sum_{i=1}^{N_s} \phi(c,s_i) = 0$. After shifting to zero mean, the sample signal variance is:

$$V_{\text{sig}} = \frac{1}{N_s}\sum_{i=1}^{N_s} \phi(c,s_i)^2.$$

If we estimated the neuron's signal variance using its responses to a single repeat, this would be upwardly biased, as it would contain a contribution from both the signal and the noise:

$$\mathbb{E}_{\nu_1}\left[\frac{1}{N_s}\sum_{i=1}^{N_s} f_1(c,s_i)^2\right] = V_{\text{sig}} + \frac{1}{N_s}\sum_{i=1}^{N_s}\mathbb{E}_{\nu}\left[\nu_1(c,s_i)^2\right],$$

where we have used the fact that the noise has zero expectation to cancel cross-terms. The upward bias introduced by noise variance could be reduced, but not eliminated, by averaging over repeats. However, we can obtain an unbiased estimate by instead computing the covariance across two repeats. Indeed, because noise has mean zero and is uncorrelated between repeats,

$$\mathbb{E}_{\nu_1,\nu_2}\left[f_1(c,s)f_2(c,s)\right] = \mathbb{E}_{\nu_1,\nu_2}\left[\left(\phi(c,s)+\nu_1(c,s)\right)\left(\phi(c,s)+\nu_2(c,s)\right)\right] = \phi(c,s)^2.$$

Thus the covariance of responses between repeats forms an unbiased estimate of the signal variance:

$$\mathbb{E}_{\nu_1,\nu_2}\left[\frac{1}{N_s}\sum_{i=1}^{N_s} f_1(c,s_i)f_2(c,s_i)\right] = V_{\text{sig}}.$$

More generally, given a population of cells $c_1 \dots c_{N_c}$, and a weight vector $\hat{\mathbf{u}} \in \mathbb{R}^{N_c}$, we can obtain an unbiased estimate of the variance of population activity projected on this dimension. Defining $\mathbf{f}_r(s) \in \mathbb{R}^{N_c}$ to be the vector summarizing these cells' responses to repeat r of stimulus s, and $\boldsymbol{\phi}(s) \in \mathbb{R}^{N_c}$ to be their noise-free responses, we can apply the same reasoning to show that:

$$\mathbb{E}_{\nu_1,\nu_2}\left[\frac{1}{N_s}\sum_{i=1}^{N_s}(\mathbf{f}_1(s_i)\cdot\hat{\mathbf{u}})(\mathbf{f}_2(s_i)\cdot\hat{\mathbf{u}})\right] = \frac{1}{N_s}\sum_{i=1}^{N_s}(\boldsymbol{\phi}(s_i)\cdot\hat{\mathbf{u}})^2$$

Thus, if the vector $\hat{\mathbf{u}}$ is an eigenvector of the stimulus-related variance, we will obtain an unbiased estimate of the corresponding stimulus-variance along this principal component.

1.1 Analysis of cvPCA method

We now consider the full cvPCA method, in which $\hat{\mathbf{u}}$ is not fixed but estimated from the data. First we will show that cvPCA finds a lower bound for the population eigenspectrum (theorem 1). Next, we will show that the expectation of the cvPCA estimate equals the actual value under conditions similar to our recordings (theorem 2).

As before, $\phi(c, s)$ represents the noise-free response of cell c to stimulus s, and $f_r(c, s) = \phi(c, s) + \nu_r(c, s)$ is its noise-corrupted response on repeat r. We consider a fixed set of cells $c_1 \ldots c_{N_c}$, and an infinite sequence of stimuli $\{s_i : i \in \mathbb{N}\}$ selected randomly from a probability distribution $\mathbb{P}(s)$. We assume that both signal and noise have finite variance: for all i, $\mathbb{E}_s\left[\phi(c_i, s)^2\right] < \infty$ and $\mathbb{E}_{s,\nu_r}\left[\nu_r(c_i, s)^2\right] < \infty$. For each repeat r, and each possible number of stimuli N_s, define the $N_s \times N_c$ data matrix $\mathbf{F}_r$ to have entries $f_r(c_i, s_k)$. We will consider the limit of applying our analysis to $\mathbf{F}_r$, as $N_s \to \infty$.

Define the signal correlation matrix $\mathbf{G}$ by $G_{i,j} = \mathbb{E}_s\left[\phi(c_i, s)\phi(c_j, s)\right]$, and the total correlation matrix $\tilde{\mathbf{G}}$ by $\tilde{G}_{i,j} = \mathbb{E}_{s,\nu}\left[f(c_i, s)f(c_j, s)\right]$, which will include contributions from both signal and noise variance. The matrices $\mathbf{G}$ and $\tilde{\mathbf{G}}$ are not random: they are statistical summaries of the (unknown) population distribution. Define the training-set sample correlation matrix $\hat{\mathbf{G}}$ as an average of the sample of N_s stimuli: $\hat{G}_{i,j} = \frac{1}{N_s}\sum_{k=1}^{N_s} f_1(c_i, s_k)f_1(c_j, s_k)$. For any finite value of N_s this is a random matrix, as the dependence on the stimulus choices and noise has not been averaged out. However, in the limit $N_s \to \infty$, it converges to the deterministic total correlation matrix $\tilde{\mathbf{G}}$ with probability 1, by the strong law of large numbers.

Let $\mathbf{u}_n$, $\hat{\mathbf{u}}_n$, and $\tilde{\mathbf{u}}_n$ be the n^{th} eigenvectors of the three correlation matrices $\mathbf{G}$, $\hat{\mathbf{G}}$, and $\tilde{\mathbf{G}}$, arranged in decreasing order of eigenvalues. Because $\lim_{N_s\to\infty} \hat{\mathbf{G}} = \tilde{\mathbf{G}}$ with probability 1, also $\hat{\mathbf{u}}_n \to \tilde{\mathbf{u}}_n$ (assuming $\tilde{\mathbf{G}}$ has no duplicate eigenvalues).

Let λ_n denote the n^{th} eigenvalue of the true signal correlation matrix $\mathbf{G}$, and $\hat{\lambda}_n$ denote the n^{th} cross-validated eigenvalue estimate:

$$
\begin{aligned}
\hat{\lambda}_n &= (\mathbf{F}_1\hat{\mathbf{u}}_n) \cdot (\mathbf{F}_2\hat{\mathbf{u}}_n)/N_s \\
&= \frac{1}{N_s}\sum_{k=1}^{N_s}\sum_{i=1}^{N_c}\sum_{j=1}^{N_c} \big(\phi(c_i, s_k) + \nu_1(c_i, s_k)\big)\big(\phi(c_j, s_k) + \nu_2(c_j, s_k)\big)\hat{u}_{n,i}\hat{u}_{n,j} \qquad (1)
\end{aligned}
$$

We are now ready to state our first theorem, which establishes the cross-validated eigenvalue estimates as a lower bound on the population signal eigenvalues:

Theorem 1. *With probability 1,*

$$
\lim_{N_s\to\infty} \mathbb{E}_{\nu_1,\nu_2}\left[\sum_{n=1}^{N}\hat{\lambda}_n\right] \leq \sum_{n=1}^{N}\lambda_n. \qquad (2)
$$

Proof. Taking the expectation of equation (1) over the noise on both repeats, we obtain

$$
\begin{aligned}
\mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right] &= \frac{1}{N_s}\sum_{k=1}^{N_s}\sum_{i=1}^{N_c}\sum_{j=1}^{N_c} \mathbb{E}_{\nu_1,\nu_2}\left[\big(\phi(c_i, s_k) + \nu_1(c_i, s_k)\big)\big(\phi(c_j, s_k) + \nu_2(c_j, s_k)\big)\hat{u}_{n,i}\hat{u}_{n,j}\right] \\
&= \frac{1}{N_s}\sum_{k=1}^{N_s}\sum_{i=1}^{N_c}\sum_{j=1}^{N_c} \mathbb{E}_{\nu_1}\left[\phi(c_i, s_k)\phi(c_j, s_k)\hat{u}_{n,i}\hat{u}_{n,j}\right] + \mathbb{E}_{\nu_1}\left[\nu_1(c_i, s_k)\phi(c_j, s_k)\hat{u}_{n,i}\hat{u}_{n,j}\right],
\end{aligned}
$$

The term ν_2 is has disappeared from the expectation because it has mean 0 and is independent of $\hat{\mathbf{u}}_n$, which is computed only from repeat 1. However, $\hat{\mathbf{u}}_n$ may be correlated with ν_1, so the expectation of the second term on the right is not necessarily zero. Nevertheless, this term will with probability 1 converge to 0 as $N_s \to \infty$. Indeed, because $\hat{\mathbf{u}}_n$ converges to the non-random quantity $\tilde{\mathbf{u}}_n$,

$$\begin{aligned}\lim_{N_s\to\infty} \mathbb{E}_{\nu_1}\left[\nu_1(c_i,s_k)\phi(c_j,s_k)\hat{u}_{n,i}\hat{u}_{n,j}\right] &= \mathbb{E}_{\nu_1}\left[\nu_1(c_i,s_k)\phi(c_j,s_k)\tilde{u}_{n,i}\tilde{u}_{n,j}\right] \\ &= \mathbb{E}_{\nu_1}\left[\nu_1(c_i,s_k)\right]\phi(c_j,s_k)\tilde{u}_{n,i}\tilde{u}_{n,j} = 0,\end{aligned}$$

since the noise has mean 0 by definition. Thus,

$$\lim_{N_s\to\infty} \mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right] = \lim_{N_s\to\infty} \frac{1}{N_s}\sum_{k=1}^{N_s}\sum_{i=1}^{N_c}\sum_{j=1}^{N_c}\phi(c_i,s_k)\phi(c_j,s_k)\tilde{u}_{n,i}\tilde{u}_{n,j} = \tilde{\mathbf{u}}_n^\top \mathbf{G}\tilde{\mathbf{u}}_n \tag{3}$$

If the vectors $\tilde{\mathbf{u}}_n$ were eigenvectors of the signal covariance matrix $\mathbf{G}$, then this last term would equal the signal eigenvalue λ_n. Instead, $\tilde{u}_n$ are the eigenfunctions of the total covariance operator $\tilde{\mathbf{G}}$. Nevertheless, as we discuss below in Theorem 2, $\mathbf{u}$ and $\tilde{\mathbf{u}}$ are likely to coincide for the current recordings. Furthermore, even if they did not coincide, we would still obtain a one-sided bound. Indeed, the Courant-Fisher minimax principle (Ref. [4], theorem 4.2.7) implies that for any set of orthonormal vectors $\tilde{\mathbf{u}}_n(c)$, the projection of $\mathbf{G}$ onto them cannot exceed what would be obtained with the true eigenvectors $\mathbf{u}_n$:

$$\sum_{n=1}^{N}\tilde{\mathbf{u}}_n^\top\mathbf{G}\tilde{\mathbf{u}}_n \leq \sum_{n=1}^{N}\mathbf{u}_n^\top\mathbf{G}\mathbf{u}_n = \sum_{n=1}^{N}\lambda_n$$

Combining this with equation (3) proves the theorem. □

We now show that the lower bound (2) will be exactly satisfied, i.e cvPCA will provide an unbiased estimate of the population eigenspectrum, when noise arises from a combination of three sources that together form an accurate description of variability in mouse visual cortex: independent variability, multiplicative scaling of population responses, and additive noise along dimensions orthogonal to the directions of signal variance [5, 6, 2]. For the current proof, we will assume that the magnitude of independent variability is equal across neurons; this assumption is not necessary in the limit that $N_c \to \infty$, but proving so will require methods to consider this limit, and is thus delayed until section 3.

Theorem 2. *Consider a noise model*

$$\nu_r(c,s) = \alpha_r(s)\phi(c,s) + \beta_r(c,s) + \gamma_r(c,s)$$

where $\alpha_r(s)$ represents multiplicative noise scaling the entire population's response to stimulus s on repeat r; $\beta_r(c,s)$ is additive noise in dimensions orthogonal to the stimulus; and $\gamma_r(c,s)$ is independent between neurons and stimuli, with α, β and γ statistically independent of each other and of $\phi(c,s)$. Then as $N_s \to \infty$, the eigenvalue estimates $\mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right]$ converge to the population eigenvalues λ_n together with additional zero eigenvalues corresponding to the additive noise dimensions.

Proof. From equation (3), we see that if an eigenvector $\tilde{\mathbf{u}}_n$ of the total covariance matrix $\tilde{\mathbf{G}}$ was equal to an eigenvector $\mathbf{u}_m$ of the signal covariance matrix $\mathbf{G}$, then $\lim_{N_s\to\infty}\mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right] = \lambda_m$. It therefore suffices to show that the eigenvectors of $\tilde{\mathbf{G}}$ are the same as those of $\mathbf{G}$, together with additional orthogonal dimensions. To do so, note that the total response

$$f_r(c,s) = (1+\alpha_r(s))\,\phi(c,s) + \beta_r(c,s) + \gamma_r(c,s)$$

Thus,

$$\begin{aligned}\tilde{G}_{i,j} &= \mathbb{E}_{s,\alpha,\beta,\gamma}\left[f_r(c_i,s)f_r(c_j,s)\right] \\ &= \mathbb{E}_{s,\alpha}\left[(1+\alpha_r(s))^2\phi(c_i,s)\phi(c_j,s)\right] + \mathbb{E}_{s,\beta}\left[\beta_r(c,s)\beta_r(c',s)\right] + \mathbb{E}_{s,\gamma}\left[\gamma_r(c,s)\gamma_r(c',s)\right].\end{aligned}$$

In matrix notation:

$$\tilde{\mathbf{G}} = V_\alpha\mathbf{G} + \mathbf{B} + V_\gamma\mathbf{I},$$

where $V_\alpha = \mathbb{E}_{\alpha,s}\left[(1+\alpha_r(s))^2\right]$ is a scale factor resulting from multiplicative modulation, $\mathbf{B}$ represents the correlation of the additive noise, $V_\gamma = \mathbb{E}_{s,\gamma}\left[\gamma_r(c_i, s)^2\right]$ is the independent noise variance, and $\mathbf{I}$ is the identity matrix.

We have assumed the additive noise dimensions to be orthogonal to the signal dimensions, so $\mathbf{B}\mathbf{u}_n = 0$. Thus, $\mathbf{u}_n$ is an eigenvector of $\tilde{\mathbf{G}}$, with eigenvalue $V_\alpha \lambda_n + V_\gamma$. Similarly, if $\mathbf{b}_m$ is an eigenvector of $\mathbf{B}$ with eigenvalue η_m, it is also an eigenvector of $\tilde{\mathbf{G}}$, with eigenvalue $\eta_m + V_\gamma$. The remaining eigenvectors of $\tilde{\mathbf{G}}$ will span the common nullspace of $\mathbf{G}$ and $\mathbf{B}$, and will all have eigenvalue V_γ.

Now, although the eigenvalues of $\mathbf{u}_n$ for the total correlation matrix $\tilde{\mathbf{G}}$ have been inflated by noise to $V_\alpha \lambda_n + V_\gamma$, this does not affect the cvPCA estimate, which is derived from the test set, and thus has expectation $\mathbf{u}_n^\top \mathbf{G} \mathbf{u}_n = \lambda_n$. In addition to the $\mathbf{u}_n$, $\tilde{\mathbf{G}}$ has a new set of eigenvectors $\mathbf{b}_m$ corresponding to the eigenfunctions of the additive noise covariance $\mathbf{B}$, and further eigenvectors spanning the common nullspace of $\mathbf{G}$ and $\mathbf{B}$. These eigenvectors are all orthogonal to the directions of signal covariance, and so are annihilated by $\mathbf{G}$. Thus, the corresponding cvPCA eigenvalue estimates will be the covariance of the training and test noise along these dimensions, which is 0 as training and test set noise is independent. Therefore, under the assumed noise model, cvPCA provides an asymptotically unbiased estimate of population eigenvalues, together with an additional set of zero eigenvalues resulting from additive noise. $\square$

2. Relating geometry to eigenspectrum decay

In this second appendix we consider the relationship between the eigenspectrum of the neural code and its geometry, and discuss the computational consequences of this relationship. Our fundamental conclusions will be that codes whose eigenspectra decay too slowly have pathological properties. We will demonstrate this by proving two theorems, and will illustrate it with three examples. The examples are counterfactual: our experimental results show that eigenspectra decay faster than $n^{-1-2/d}$, and the examples illustrate the problems that would occur if eigenspectra decayed more slowly than this.

We consider here only the geometry of noise-free responses: as shown in Appendix 1, the cvPCA method provides an unbiased estimate of the noise-free eigenspectrum. We consider their geometry in the limit as a large number of neurons are recorded, which allows us to consider neural activity in an infinite-dimensional coding space. The mathematical technicalities and philosophy of this many-neuron limit will be discussed in more detail in Appendix 3.

2.1 Why can't eigenspectra be flat?

Describing our main results will require building up some mathematical machinery concerning fractal dimensions. However, even before doing this it is possible to gain an intuition for the pathology of slowly-decaying eigenspectra, by considering a simple counterfactual example.

Example 1. This first example is the efficient coding hypothesis *in extremis*: all neurons encode the stimulus completely independent of each other. There are no signal correlations, and the eigenspectrum is therefore flat. Although neurons certainly could display independent noise, this example shows why independent encoding of a signal is pathological in large populations.

In this example, the stimulus to be encoded is a single number, and neurons encode its binary representation. Let the stimulus s be uniformly distributed between 0 and 1, and let cell c_i encode the i^{th} digit in the binary expansion of s. Because s is uniformly distributed, all neurons are independent, and the eigenspectrum is flat. Real neurons, of course, do not form binary codes: however, by considering this simple counterfactual case, we will gain an insight into the pathological properties any independent code would also have to exhibit.

To understand the pathologies of this code, suppose the stimulus was represented by a population of 1,000 neurons. The first ten cells would represent the stimulus to an accuracy of one part in 1024, which would almost certainly exceed the accuracy to which the stimulus could be measured by sensory systems. The remaining 990 cells would represent the stimulus to a supposed accuracy of more than one part in 10^{300}. Nearly all the variance of the population would therefore be used to encode absurdly fine details of the stimulus. A downstream population could only recover useful information about the stimulus by "finding a needle in a haystack," i.e. by basing its responses on just these 10 neurons, while ignoring all others. If the downstream structure received functional inputs from anything but this tiny fraction of cells, then a minute change in the stimulus would cause it to receive an almost completely different input, meaning that no generalization would be possible.

The code just described is highly susceptible to neural noise, but that is not its fundamental pathology. If the population contained many neurons redundantly encoding each digit in the binary expansion, a downstream structure could recover the signal accurately despite noise by averaging their activity. Still however, 0.1% of the population would encode the stimulus to an accuracy of over one part in 1000, with the remaining 99.9% of neurons encoding irrelevant details.

The same pathology would occur for higher dimensional stimuli. Indeed, consider the example of N completely independent binary neurons encoding a stimulus described by d real variables. The first $10d$ neurons would encode these variables to an accuracy of 1 in 1024, with the remainder of the population again representing features that could not in reality be measured by sensory systems. If $d \ll N$, then nearly all the population variance is devoted to encoding irrelevant stimulus details. Fully uncorrelated stimulus encoding is therefore pathological whenever the number of neurons substantially exceeds the dimension of the stimulus being encoded.

We have presented an example of binary neurons, but it is possible to extend it to an (even more counterfactual) case of continuous firing rates, illustrating the types of pathology that we will prove must happen for any codes whose eigenspectra decay too slowly. Considering again a one-dimensional stimulus, we can obtain a continuous code by smoothly joining the 2^N possible binary response vectors into a smooth curve. Because any two such vectors are separated by a distance of at least 1 in neural space, this curve has length at least 2^N. The average derivative of the neural response vector is the length of the neural manifold divided by the length of the stimulus manifold, which is independent of N. Thus, the derivative scales exponentially with N, and a tiny change in the stimulus would cause a huge change in the population response, reflecting a code which cannot generalize between stimuli.

In the remainder of this section, we will prove that the type of pathologies illustrated by this example are inevitable if eigenspectra do not decay sufficiently rapidly. The key to this argument is that later principal components encode finer details of the stimulus. This is empirically true in our data (Extended Data Fig. 6) and is also a mathematical necessity. The number of slowly-varying orthogonal functions of the stimulus is limited (assuming a finite-dimensional, smooth, and compact stimulus space). Because the principal components are orthogonal, they must therefore asymptotically encode ever finer stimulus details. If the eigenspectrum decays too slowly, population activity will be overwhelmed by representation of these ever finer details, resulting in a code with the same pathologies as the binary example.

2.2 Multiple notions of dimensionality

A key concept in this work is the dimensionality of a neural representation. We will require four different notions of dimensionality, which in general are not equal.

The first notion is *ambient dimension*. By this we mean the dimension of a vector space, i.e. the number of coordinates required to identify a point. For example, we can summarize the responses of N_c cells to a stimulus s by a vector in an N_c-dimensional vector space, which we refer to as the *ambient space*.

The second notion is *planar dimension*. Not every point in the ambient space needs to be a response to a stimulus; indeed, as there are more neurons in V1 than fibers in the optic nerve, the actual response patterns evoked by stimuli must occupy a subset of the ambient space. We define the planar dimension of a neural code to be the dimension of the smallest vector subspace containing all firing patterns that can actually be produced. Thus, if the planar dimension is n, then the response to any stimulus s can be represented as a sum of n basis vectors: $\boldsymbol{\phi}_s = \sum_{i=1}^{n} \phi_i \mathbf{u}_i$.

The third notion is *manifold dimension*. A *manifold* is a generalization of a continuous surface to arbitrary dimensionality. In any local region of a d-dimensional manifold, each point can be uniquely identified by d coordinates. However, unlike a vector space, a manifold may be curved, and the functions relating these coordinates to the points on the manifold can be nonlinear. A manifold can be embedded in a vector space of equal or higher ambient dimension than the manifold dimension. For example, the earth's surface defines a 2-dimensional manifold embedded in a 3-dimensional ambient space.

Our final notion of dimensionality is *fractal dimension*. Fractal dimension measures the roughness of a geometric object. For example, the coastline of a country has manifold dimension 1, but can be "rough" in the sense that the higher the resolution, the more complicated and longer the coastline appears to be. Precise definitions of fractal dimension are based on the idea that the volume of a d-dimensional object with diameter δ should scale as δ^d. The west coast of Britain, for example, has a fractal dimension of approximately 1.25 (Ref. [7]), meaning that the number of times a ruler of length δ can be laid end-to-end around the coastline scales as $\delta^{-1.25}$. A *fractal* is an object whose fractal dimension exceeds its manifold dimension. Smooth manifolds, whose coordinate functions are differentiable, cannot be fractals.

There are multiple subtly-differing ways to formalize the fractal dimension [8, 9]. Here we will use the *upper Minkowski dimension*, which is defined by counting the number of spheres of diameter δ required to cover the object. If we call this number N_δ, the Minkowski dimension is essentially the limit of $\log(N_\delta)/\log(\delta^{-1})$ as $\delta \to 0$ (see section 2.6 for precise definition).

2.3 Summary of results

We are now able to summarize our mathematical results relating the geometry of the neural code to its eigenspectrum. These results demonstrate that unless eigenspectra decay faster than n^{-1}, population codes are pathological, either exhibiting discontinuous responses, or infinite population variance. Furthermore, for stimuli drawn from a set of manifold dimension d, codes with eigenspectra decaying slower than $n^{-1-2/d}$ are also pathological, displaying infinite variance of the code's derivative and fractal geometry of the response manifold. We conclude that our experimental observations of eigenspectrum decay only just faster than $n^{-1-2/d}$ indicate a neural code that is as high-dimensional as possible before hitting the regime where these pathological conditions must occur. More detailed descriptions of each result now follow.

Theorem 3 demonstrates the pathology of neural population codes whose eigenspectra do not decay faster than n^{-1}. We show that if the sum of the population eigenvalues is infinite, then either the neural code is discontinuous or population activity has infinite variance. Because $\sum_{n=1}^{\infty} n^{-\alpha} = \infty$ when $\alpha \leq 1$, a code whose eigenspectrum did not decay faster than n^{-1} must exhibit one of these properties. Both possibilities render a neural code pathological, resulting respectively in complete failure to generalize, or requiring neurons to respond to preferred stimuli with arbitrarily large firing rates.

Theorem 4 illustrates the pathological features of eigenspectra decaying more slowly than $n^{-1-2/d}$. This theorem shows that if the fractal dimension of an object is d, then its covariance eigenspectrum cannot decay slower than $n^{-1-2/d}$. Applying this to the set of neural population responses to a stimulus ensemble of manifold dimension d, we see that if its eigenspectrum decayed slower than $n^{-1-2/d}$, the responses must define a fractal in the ambient space. The "rough" nature of such population codes means they are unsuitable for representing differences between similar stimuli: we prove in Corollary 4.1 that any map from a d-dimensional stimulus manifold to a space of fractal dimension $> d$ cannot be differentiable, so there is no way a population whose eigenspectrum decayed this slowly could smoothly encode a d-dimensional stimulus. The precise form of this non-differentiability is made explicit in the next theorem.

Theorem 5 relates the derivative of the neural population representation with respect to a stimulus of manifold dimension d, to its eigenspectrum. We prove that later eigenfunctions must encode finer stimulus details, and specifically that the derivative magnitude of the n^{th} eigenfunction must grow as order n^{2d}. This implies that unless the eigenspectrum decays faster than $n^{-1-2/d}$, the derivative of the population response vector has infinite expected magnitude. The pathology of non-differentiable population codes occurs because the difference in population response to two close stimuli is proportional to the code's derivative. If the derivative is infinite, then the population codes for extremely similar stimuli can still be very different, resulting in failure to generalize responses to arbitrarily close stimuli. This theorem also shows that the smoothest codes possible for a given eigenspectrum will have principal components whose dependence on the stimulus is given by Laplacian eigenfunctions, as would be found by the Laplacian eigenmap algorithm often used for unsupervised learning [10].

Example 2 illustrates theorems 4 and 5 with a representation of stimulus of manifold dimension 1 in an infinite-dimensional ambient space. Consistent with theorem 4, we show that if the eigenspectrum decays slower than n^{-3} then the set of possible responses is fractal. Consistent with theorem 5, we show that the derivative is infinite if the eigenspectrum decays as n^{-3} or slower. Though analytically tractable, this representation takes the form of coordinates in an abstract Hilbert space, rather than explicitly-modelled neurons.

In example 3 we consider a more explicitly neural representation, built from units with Gaussian radial basis function receptive fields over an abstract stimulus space. We show that if all units have equal tuning radius, the eigenspectrum does not follow a power law but is approximately flat until the eigenfunctions' spatial scale matches this radius, after which the eigenspectrum falls rapidly. We then show how a scale-free mixture of units of varying radii results in a power-law eigenspectrum. By varying the balance of broadly and sharply tuned neurons, we show that the population code is not differentiable if its eigenspectrum decays slower than $n^{-1-2/d}$. We show that in this case the neural representation of the derivative is dominated by increasingly rare neurons of increasingly sharp tuning, resulting in an infinite population gradient with respect to the stimulus, even though every neuron's individual response is an infinitely-differentiable Gaussian function.

2.4 Relation to kernel learning theory

To consider the computational consequences of different cortical coding geometries, one should consider how these geometries affect the ability of downstream structures to form associations between sensory stimuli and appropriate behaviors. This reveals an intriguing parallel between our findings and the theory of kernel machines, a class of machine learning algorithms that provide a useful analogy to describe features of cortical coding [11, 12]. We will discuss this parallel using a model where the cortical code is fixed, and downstream structures learn stimulus-behavior associations through Hebbian mechanisms such as the delta rule [13], leading to a behavioral output determined by nonlinear function of a linear projection of cortical population activity (linear-nonlinear function). Neither our results nor their interpretation rely on this assumption, but it provides a simple way to illustrate the computational consequences of different representational geometries.

The way animals generalize responses to sensory stimuli will depend on the overlap of the corresponding population codes. Suppose an animal had learned to produce a behavioral response to a sensory stimulus s. If the animal later experienced another stimulus s' that evoked a similar pattern of cortical population activity, this stimulus would likely produce the same behavioral response, even if the animal had never experienced s' before. However, if s and s' activate different cortical populations, such generalization would be unlikely to occur.

Sensory inputs that generate strong responses in a large number of cortical cells will be salient, meaning that the animal is likely to rapidly learn responses to them. Indeed, if downstream structures learn by Hebbian plasticity, then pairing a training signal with a stimulus that strongly drives many cortical cells will strengthen many synapses, which will strongly drive downstream neurons when the stimulus next appears. However pairing the training signal with a stimulus that weakly drives only a few cortical cells would only strengthen a few synapses, and so lead to only a weak downstream response.

These intuitive notions can be formalized in geometric terms. The geometrical relationship between the population vectors $\boldsymbol{\phi}_s$ and $\boldsymbol{\phi}_{s'}$ evoked by two stimuli can be summarized by their inner product $\langle \boldsymbol{\phi}_s, \boldsymbol{\phi}_{s'} \rangle$, which is termed the *kernel function* $K(s, s')$. The total amount of population activity evoked by stimulus s can be summarized by the length of the population vector, $\|\boldsymbol{\phi}_s\|$, which is given by $\sqrt{K(s,s)}$. The similarity of the responses to s and s' can be captured by the correlation coefficient, which is equal to $K(s,s')/\sqrt{K(s,s)K(s',s')}$. Thus, not only is the geometry of the cortical representation critical to downstream Hebbian learning, but the important features of this geometry are captured by the kernel function.

The kernel function summarizes the similarity of cortical responses to two stimuli, rather than the physical similarity of the stimuli themselves (i.e. the patterns of light falling on the retina). For example, sensory processing might render the population responses to physically similar but behaviorally different stimuli (such as an edible and a poisonous mushroom) as orthogonal, but cause two physically different stimuli of similar behavioral relevance (such as different instantiations of a similar visual texture) to have a similar population representation.

To understand how the kernel function determines downstream learning, it is useful to adopt the "function space" description of kernel learning systems [14, 15]. The kernel function is defined not just for the particular sample of stimuli shown in an experiment, but over the entire set of possible stimuli that the animal might potentially encounter. This function represents an infinite-dimensional generalization of a matrix, and we can define its eigenspectrum, although because the set of potential stimuli is infinite, there may now be an infinite number of eigenvalues. Indeed (given a technical condition known as compactness) the kernel function has a set of eigenfunctions $v_n(s)$ and real non-negative eigenvalues λ_n such that $\mathbb{E}_{s'}\left[K(s,s')v_n(s')\right] = \lambda_n v_n(s)$ (Ref. [4], Theorem 4.2.4). This definition, based on the expectation over a continuous random variable, generalizes the finite-dimensional notion of eigenvectors based on matrix multiplication. Similarly to how finite-dimensional PCA produces orthonormal eigenvectors, these eigenfunctions are uncorrelated and have unit variance: $\mathbb{E}_s\left[v_n(s)v_m(s)\right] = \delta_{n,m}$.

We can formalize a response that the system might need to learn as a function $f(s)$, again defined over the infinite set of all stimuli that might potentially be encountered. If downstream learning occurs via a delta rule, the dynamics by which this function will be learned from examples is determined by the kernel eigenspectrum. The system will first learn the projection of f onto the eigenfunctions of large eigenvalues, corresponding to the dimensions of maximum variance of the neural representation. The projection of f onto eigenfunctions with smaller eigenvalues will be added to the system's output only after more extensive training, at a speed proportional to the eigenvalues [16, 17, 18].

The kernel eigenspectrum therefore determines the flexibility of the set of functions the system is able to learn. If the eigenspectrum decays rapidly, associations will be biased towards functions expressible as combinations of those few eigenfunctions with large eigenvalues. Because fine-scale stimulus features are represented in eigenfunctions with small eigenvalues (shown in theorem 5 below), rapid eigenspectrum decay means that a crude representation will be learned quickly, but fine details cannot be distinguished without extensive training. If the eigenspectrum decays slowly, the system is able to learn much more detailed associations, but this comes at a price: the system is less able generalize from one stimulus to the next, as the representation of these stimuli is more likely to be orthogonal. Thus, the rate of eigenspectrum decay can be seen as a smoothing parameter: slower eigenspectrum decay means less smoothing across stimuli, which enables representation of fine stimulus features but impairs generalization.

This viewpoint provides a new way of understanding the n^{-1} and $n^{-1-2/d}$ bounds. If the eigenspectrum decayed slower than n^{-1}, then the representation of fine details would overwhelm the representation of broad stimulus features: because $\sum_{n=1}^{\infty} n^{-1}$ does not converge, the leading eigenfunctions will make up an infinitesimal fraction of the total neural variance, the functions the system learned would be dominated by fine details, and convergence to the correct function f would not occur after any finite training time. If the eigenspectrum decayed slower than $n^{-1-2/d}$, the functions learned by the system would fail to be differentiable, and responses to one stimulus need not generalize to other nearby stimuli.

2.5 Implications of n^{-1} eigenspectra

We now move to a formal characterization of the pathologies that occur if eigenspectra decay too slowly. To a reader interested in learning more of the relevant background material, we recommend Ref. [4], which covers almost all the required material on probability in infinite-dimensional spaces, and Ref. [9] which covers all the required material on fractal geometry. For applications of similar mathematics in machine learning, we recommend refs. [19, 20, 21], and for a more general background in functional analysis, refs. [22, 23]. The key material on Riemannian and spectral geometry can be found in Ref. [24].

Our first theorem concerns the pathologies of neural representations whose eigenspectra decay slower than $1/n$. We show that if the population eigenspectrum decays slower than $1/n$, then either the kernel function is discontinuous, or the variance of the population is infinite. The former case would mean that the neural code cannot generalize responses to new stimuli. The latter would mean that population responses are dominated by rare neurons responding at arbitrarily high rates to certain stimuli. From a mathematical perspective, this theorem follows immediately from basic results; this subsection will therefore be devoted to introducing the terminology required to describe why.

To characterize the geometry of the population code in the limit of a large number of cells, we must consider activity to reside in an infinite-dimensional vector space. For finite neural populations, we can define the distance between responses to two stimuli using Euclidean distance: if the vectors $\boldsymbol{\phi}_s$ and $\boldsymbol{\phi}_{s'}$ contain the responses of cells $\{c_1, c_2, ..., c_{N_c}\}$ to stimuli s and s', we define the distance between them as $\|\boldsymbol{\phi}_s - \boldsymbol{\phi}_{s'}\|^2 = \frac{1}{N_c}\sum_{i=1}^{N_c}(\phi(c_i,s) - \phi(c_i,s'))^2$. The normalization factor $\frac{1}{N_c}$ allows us to take a limit as the number of cells increases, as described in more detail in section 3. We define the distance in this limit as the expected squared difference in the firing rate of a randomly chosen neuron: $\|\boldsymbol{\phi}_s - \boldsymbol{\phi}_{s'}\|^2 = \mathbb{E}_c\left[(\phi(c,s) - \phi(c,s'))^2\right]$. This distance makes the space of responses into a *Hilbert space*, a type of infinite-dimensional space with many properties similar to finite-dimensional Euclidean spaces. Hilbert spaces, like finite-dimensional spaces, have an inner product, $\langle \boldsymbol{\phi}_s, \boldsymbol{\phi}_{s'} \rangle = \mathbb{E}_c\left[\phi(c,s)\phi(c,s')\right]$. Like finite-dimensional vector spaces, they can be given an orthonormal basis, although now there may be an infinite number of basis vectors. A vector in a Hilbert space can be expressed as an infinite linear combination of basis vectors: $\boldsymbol{\phi} = \sum_{i=1}^{\infty} \phi_i \mathbf{u}_i$, and Pythagoras' theorem holds: $\|\boldsymbol{\phi}\|^2 = \sum_{i=1}^{\infty} \phi_i^2$. Unlike the finite-dimensional case, however not every set of coordinates defines a vector: a vector only belongs to the Hilbert space if it has finite length, $\sum_{i=1}^{\infty} \phi_i^2 < \infty$.

Recall that the *trace* of a finite-dimensional matrix is the sum of its diagonal elements. A standard theorem of linear algebra holds that the trace of a matrix equals the sum of its eigenvalues (e.g. Ref. [25], pp. 336). Traces can also be defined in infinite-dimensional Hilbert spaces. The infinite-dimensional equivalent of a matrix is an *operator*: a function that linearly maps infinite-dimensional vectors to infinite-dimensional vectors. For our current application, the most important operators are *integral operators*. For example, given a probability space $\mathcal{S}$ of possible stimuli, the space of all functions $a(s)$ with finite squared expectation ($\mathbb{E}_s\left[a(s)^2\right] < \infty$) defines an infinite-dimensional Hilbert space termed $L^2(\mathcal{S})$. The kernel function $K(s, s')$ defines an operator $\mathscr{K}$ on $L^2(\mathcal{S})$, as the linear mapping that takes a function $a(s)$ to the function $\mathscr{K} a(s) = \mathbb{E}_{s'}\left[K(s', s)a(s')\right]$. (It is called an integral operator because the expectation can be expressed as an integral over s'.) The trace of the operator $\mathscr{K}$ is defined as the sum of its diagonal elements with respect to any orthonormal basis e_i: $\mathrm{Tr}\,(\mathscr{K}) = \sum_i \mathbb{E}_s\left[e_i(s)\mathscr{K} e_i(s)\right]$. As in finite dimensions, the trace of an operator is equal to the sum of its eigenvalues: $\mathrm{Tr}\,(\mathscr{K}) = \sum_i \lambda_i$. However, unlike in finite dimensions, the trace need not be finite; there are an infinite number of eigenvalues, which might not have a finite sum. An operator whose trace is finite is called a *trace class* operator (Ref. [4], section 4.5). For integral operators with a continuous kernel, the trace can be expressed as an expectation along the diagonal (Ref. [4] theorem 4.6.7; Ref. [26], lemma to theorem XI.31):

$$\mathrm{Tr}\,(\mathscr{K}) = \mathbb{E}_s\left[K(s, s)\right].$$

Because $K(s, s) = \mathbb{E}_c\left[\phi(c, s)^2\right]$, this establishes:

Theorem 3. *Let $\lambda_1 \geq \lambda_2 \geq \ldots \geq 0$ be the eigenspectrum for a population code $\phi(c, s)$ whose kernel function $K(s, s')$ is continuous. Then*

$$\sum_{i=1}^{\infty} \lambda_i = \mathbb{E}_{s,c}\left[\phi(c, s)^2\right].$$

In particular, the eigenvalue sum is finite if and only if the population variance is finite. □

The relevance of this to eigenspectrum decay comes from the fact that $\sum_{n=1}^{\infty} n^{-1}$ is infinite. Thus, a continuous kernel and finite population variance means that the eigenspectrum must decay faster than this. Formally:

Corollary 3.1. *If $\mathbb{E}_{s,c}\left[\phi(c, s)^2\right]$ is finite and K is continuous then $\lambda_n = o(n^{-1})$.*

Proof. Recall that $\lambda_n = o(n^{-1})$ means that for any $\epsilon > 0$ there exists N such that $\lambda_n \leq \epsilon n^{-1}$ for all $n \geq N$. Theorem 3 shows $\sum_{i=1}^{\infty} \lambda_i$ is finite, thus for any ϵ there exists N such that $\sum_{i=n/2}^{\infty} \lambda_i \leq \epsilon/2$ for all $n \geq N$, where $n/2$ is understood to be rounded down for odd n. Because λ_n is non-negative and non-increasing, $n\lambda_n/2 \leq \sum_{i=n/2}^{n} \lambda_i \leq \sum_{i=n/2}^{\infty} \lambda_i \leq \epsilon/2$. Thus, $\lambda_n \leq \epsilon n^{-1}$ for all $n \geq N$. □

Thus, a code whose eigenspectrum decayed more slowly than n^{-1} would be pathological in one of two ways. The first way, a discontinuous kernel function, would mean no generalization: the response of the population to neighboring stimuli can be completely different, no matter how close these stimuli are. The completely uncorrelated code described in example 1 was an example of a code with discontinuous kernel function; we will make this statement mathematically precise in section 3.1. The second way, infinite population variance would mean neurons respond to preferred stimuli at arbitarily high rates. We will provide an example of this other form of pathological code as example 5.

2.6 Relating fractal dimension to covariance eigenvalues

The previous section showed that population codes with eigenspectra decaying slower than n^{-1} are highly pathological, exhibiting either kernel discontinuity or infinite variance. However, the potential pathologies of neural codes are not restricted to these extreme problems. A neural population code can exhibit a more subtle form of pathology by failing to be *differentiable*. Functions can be continuous but not differentiable: for example, the function $y = \sqrt[3]{x}$ is continuous but not differentiable at $x = 0$, as the curve has an infinite slope at this point. Furthermore, there are functions that are continuous everywhere but differentiable nowhere [27]. While such non-differentiable functions are surprising, they can be useful models of some phenomena occurring in nature. Indeed, because fractals cannot have differentiable coordinate functions, objects such as coastlines can be modeled by functions that are continuous but not differentiable.

In infinite dimensions, the concept of differentiability has an extra subtlety. Consider a vector $\boldsymbol{\phi}(x) = \sum_{i=1}^{\infty} \phi_i(x)\mathbf{u}_i$, in a Hilbert space with orthonormal basis $\{\mathbf{u}_i\}$, depending on a scalar parameter x. As in finite dimensions, we can formally express the derivative of the vector as $\frac{d\boldsymbol{\phi}}{dx} = \sum_{i=1}^{\infty} \frac{d\phi_i}{dx}\mathbf{u}_i$. In a finite-dimensional space, $\boldsymbol{\phi}(x)$ will be differentiable if all of its coordinates $\phi_i(x)$ are differentiable. But in infinite dimensions this need not be the case: $\boldsymbol{\phi}(x)$ can fail to be differentiable even if every coordinate function is differentiable, if the infinite sum $\sum_{i=1}^{\infty} \left|\frac{d\phi_i}{dx}\right|^2$ does not converge.

We now prove two theorems characterizing the relationship between eigenspectrum decay and differentiability of the population code. In the first we prove that the eigenspectrum of neuronal responses constrained to a set of fractal dimension d must decay at least as fast as a power law of exponent $\alpha = 1 + 2/d$. This means that if the eigenspectrum of a code of manifold dimension d decays slower than this, the response set must be fractal.

There are several ways of formalizing the notion of fractal dimension (See e.g. Refs. [8, 9]), all based on the idea that the volume of a d-dimensional object with diameter δ should scale as δ^d. Here, we will use the *upper Minkowski dimension*. Given a subset $\mathcal{F}$ of a metric space, we define the the *covering number* $N_\delta(\mathcal{F})$ to be the smallest number of spheres of diameter δ required to cover $\mathcal{F}$. Intuitively, $N_\delta(\mathcal{F})$ should scale as δ^{-d} for a space of dimension d. The upper Minkowski dimension of $\mathcal{F}$ is written as $\overline{\dim}_M\mathcal{F}$, and defined to be $\inf\{d' : \limsup_{\delta\to 0} N_\delta(\mathcal{F})\delta^{d'} = 0\}$. This means that for any $d' > \overline{\dim}_M\mathcal{F}$ and for any $C > 0$, there exists an $\epsilon > 0$ such that for all $\delta < \epsilon$, $N_\delta(\mathcal{F})\delta^{d'} < C$; furthermore $\overline{\dim}_M\mathcal{F}$ is the smallest number with this property. We can now prove our theorem.

Theorem 4. *Let $\boldsymbol{\phi}(s)$ be a function of a random element s, supported on a set $\mathcal{F}$ of upper Minkowski dimension d inside a separable Hilbert space $\mathbb{H}$, with $\mathbb{E}_s\left[\|\boldsymbol{\phi}(s)\|^2\right] < \infty$. Write the eigenvalues of Cov$(\boldsymbol{\phi}(s))$ as $\lambda_1 \geq \lambda_2 \geq \ldots$. Then for all $d' > d$, $\lambda_n = O(n^{-1-2/d'})$ as $n \to \infty$.*

Proof. We assume without loss of generality that $\mathbb{E}_s[\boldsymbol{\phi}(s)] = 0$. Fix $d' > \overline{\dim}_M\mathcal{F}$ and $C > 0$. By the defintion of upper Minkowski dimension, there exists n_0 such that for any $n \geq n_0$, we can cover $\mathcal{F}$ with n balls of diameter at most $Cn^{-1/d'}$. For each s, let b_s be an integer between 1 and n identifying a ball in this cover that contains $\boldsymbol{\phi}(s)$. Define a mean vector for each ball: $\boldsymbol{\mu}_s = \mathbb{E}_{s'}[\boldsymbol{\phi}(s')|b_{s'} = b_s]$. We can decompose the variance of $\boldsymbol{\phi}(s)$ using an ANOVA decomposition:

$$\mathbb{E}_s\left[\|\boldsymbol{\phi}(s)\|^2\right] = \mathbb{E}_s\left[\|\boldsymbol{\phi}(s) - \boldsymbol{\mu}_s\|^2\right] + \mathbb{E}_s\left[\|\boldsymbol{\mu}_s\|^2\right]. \tag{4}$$

Let $\mathscr{P}$ be the projection operator onto the $\leq n$-dimensional subspace spanned by mean vectors $\{\boldsymbol{\mu}_s\}$. Then the total variance of $\boldsymbol{\phi}(s)$ in this subspace is at least as big as the variance between means:

$$\mathbb{E}_s\left[\|\mathscr{P}\boldsymbol{\phi}(s)\|^2\right] = \mathbb{E}_s\left[\|\mathscr{P}(\boldsymbol{\phi}(s) - \boldsymbol{\mu}_s)\|^2\right] + \mathbb{E}_s\left[\|\boldsymbol{\mu}_s\|^2\right] \geq \mathbb{E}_s\left[\|\boldsymbol{\mu}_s\|^2\right].$$

The variance in any n-dimensional subspace cannot exceed the sum of the first n covariance eigenvalues: $\mathbb{E}_s\left[\|\mathscr{P}\boldsymbol{\phi}(s)\|^2\right] \leq \sum_{i=1}^{n} \lambda_i$ (e.g. Ref. [4], theorem 7.2.8). Thus,

$$\sum_{i=1}^{n} \lambda_i \geq \mathbb{E}_s\left[\|\boldsymbol{\mu}_s\|^2\right]. \tag{5}$$

Write the sum of all eigenvalues above n as $\Lambda_n = \sum_{i>n} \lambda_i$. Because the eigenvalues must sum to the total variance, $\sum_{i=1}^{\infty} \lambda_i = \mathbb{E}_s\left[\|\boldsymbol{\phi}(s)\|^2\right]$, combining (4) and (5) gives $\Lambda_n \leq \mathbb{E}_s\left[\|\boldsymbol{\phi}(s) - \boldsymbol{\mu}_s\|^2\right]$. Furthermore, because the balls have diameter at most $Cn^{-1/d'}$, this implies that for all $n \geq n_0$,

$$\Lambda_n \leq C^2 n^{-2/d'}$$

Informally we can see why λ_n should follow an $n^{-1-2/d'}$ power law by differentiating with respect to n. Formally, consider any even $n > 2n_0$. Then $\Lambda_{n/2} = \sum_{i=n/2+1}^{\infty} \lambda_i \geq \sum_{i=n/2+1}^{n} \lambda_i \geq \frac{n}{2}\lambda_n$, as the λ_i are positive and non-increasing. Thus $\lambda_n \leq \frac{2}{n}\Lambda_{n/2} \leq \frac{2}{n}C^2(\frac{n}{2})^{-2/d'} = 2^{1+2/d'}C^2 n^{-1-2/d'}$. Because C was arbitrary, $\lambda_n = O(n^{-1-2/d'})$. □

We now prove a corollary to this theorem, which explains its relevance to stimulus encoding. Suppose a stimulus is drawn from a manifold of dimension d, and mapped to the neural space by a code with bounded derivative (which means it satisfies a property of Lipschitz continuity). The corollary shows the response set must have fractal dimension $\leq d$, and thus an eigenspectrum decaying faster than $n^{-1-2/d'}$ for any $d' > d$. Thus, even though theorem 4 only speaks to the geometry of the response set, it has implications for the neural coding of sensory stimuli. These implications will be discussed further in the next section.

Corollary 4.1. *If $\mathcal{F} \subset \mathbb{H}$ is a the image of a set of upper Minkowski dimension d under a Lipschitz map, or if $\mathcal{F}$ is a d-dimensional compact differentiable submanifold of $\mathbb{H}$, then for all $\mathrm{d}' > d$, $\lambda_n \sim O(n^{-1-2/d'})$.*

Proof. In both cases we must show that $\mathcal{F}$ has upper Minkowski dimension at most d. It is a standard result that if f is a Lipschitz map, $\overline{\dim}_M(f(\mathcal{F})) \leq \overline{\dim}_M(\mathcal{F})$, and that if f is bi-Lipschitz, $\overline{\dim}_M(f(\mathcal{F})) = \overline{\dim}_M(\mathcal{F})$ (Ref. [9], Prop. 2.5).

By a d-dimensional differentiable submanifold of $\mathbb{H}$ we mean a set $\mathcal{F} \subset \mathbb{H}$, where every point is contained in an open set (of the subspace topology on $\mathcal{F}$) that bijects to an open set of $\mathbb{R}^d$ via a chart with a continuous Frechet derivative that is nonzero everywhere. To show such a set has upper Minkowski dimension d, observe that each chart must be bi-Lipschitz, and thus each point is contained in an open set of upper Minkowski dimension d. By compactness, $\mathcal{F}$ is therefore covered by finite number of sets of upper Minkowski dimension d, and because $\overline{\dim}_M(\mathcal{F}_1 \cup \mathcal{F}_2) = \max(\overline{\dim}_M \mathcal{F}_1, \overline{\dim}_M \mathcal{F}_2)$ (Ref. [9], p.35), $\overline{\dim}_M \mathcal{F} = d$. □

2.7 Insufficient decay means infinite derivative

To gain additional insight into the $n^{-1-2/d}$ bound, we now explicitly show how if eigenvalues did not decay this fast, the expected squared derivative of the neural code would have to be infinite. To do so, we consider the space of stimuli $\mathcal{S}$ to be a Riemannian manifold. A Riemannian manifold has more structure than a differentiable manifold: a differentiable manifold allows one to compute a derivative, but a Riemannian manifold also allows one to measure its magnitude. We will make use of three theorems from Riemannian geometry. The first of these, Green's theorem (Ref. [24], theorem III.10), provides an estimate of the magnitude of the derivative of a smooth function on a Riemannian manifold. If $f(s)$ is a smooth and and compactly supported function on a Riemannian manifold, Green's theorem implies

$$\int |\nabla f(s)|^2 d\mu(s) = \int f(s) \triangle f(s) d\mu(s).$$

Here, $|\nabla f(s)|^2$ represents the squared magnitude of the derivative of f with regards to the Riemannian metric, and $\int d\mu(s)$ represents integration on the manifold using the canonical measure defined by the Riemannian metric. $\triangle$ represents the Laplace-Beltrami operator, a generalization of the Laplacian operator $-\sum_i \frac{\partial^2}{\partial x_i^2}$ familiar from physics in Euclidean spaces, to the case of general Riemannian manifolds.

The second standard result we use concerns the eigenvalues of the Laplace-Beltrami operator. The operator has a set of eigenfunctions $w_n, n \in \mathbb{N}$, with corresponding eigenvalues η_n, such that

$$\triangle w_n(s) = \eta_n w_n(s)$$

The eigenfunctions w_n form a basis for $L^2(\mathcal{S})$, the Hilbert space of square-integrable functions on the manifold, and the eigenvalues $0 \leq \eta_1 \leq \eta_2 \ldots$ are non-negative, increasing, and unbounded (Ref. [24], theorem III.18).

The third result we require is Weyl's asymptotic law (Ref. [24], theorem III.36), which relates the growth of the Laplace-Beltrami eigenvalues to the dimension of the manifold. This law states that for a mainfold of dimension d,

$$\eta_n \sim n^{2/d}$$

In other words, there exists a constant $C_{\mathcal{S}}$ such that $\lim_{n\to\infty} \eta_n n^{-2/d} = C_{\mathcal{S}}$. (The value of $C_{\mathcal{S}}$ can be computed using the total volume of the manifold, but its precise value does not concern us here.) This is a remarkable result, as it means the asymptotic distribution of larger eigenvalues does not depend on the shape of the manifold, only its volume and dimensionality.

With these preliminaries, we are now ready to prove our next theorem. We consider stimuli to be drawn at random from a closed, compact d-dimensional Riemannian manifold $\mathcal{S}$ according to its canonical Riemannian measure. We assume that the responses have finite squared expectation: $\mathbb{E}_s\left[\|\boldsymbol{\phi}(s)\|^2\right] < \infty$. This activity defines a kernel function $K(s,s') = \langle \boldsymbol{\phi}(s), \boldsymbol{\phi}(s)\rangle$, with positive bounded eigenvalues λ_n, arranged in decreasing order: $\lambda_1 \geq \lambda_2 \geq \ldots \geq 0$. Then we have

Theorem 5. *If the expected derivative magnitude of $\boldsymbol{\phi}$ is finite, i.e. $\mathbb{E}_s\left[\|\nabla_s \boldsymbol{\phi}(s)\|^2\right] < \infty$, then $\lambda_n = o(n^{-1-2/d})$.*

Proof. Our proof rests on relating the kernel eigenfunctions $v_i(s)$ to the Laplacian eigenfunctions $w_i(s)$. Since they are both orthonormal bases for $L^2(\mathcal{S})$, we may define a set of orthonormal matrix elements $\{a_{i,j}\}$ such that $v_i = \sum_j a_{i,j} w_j$ and $w_j = \sum_i a_{i,j} v_i$, with $\sum_i a_{i,j} a_{i,k} = \delta_{j,k}$, and $\sum_j a_{i,j} a_{k,j} = \delta_{i,k}$.

Because the kernel eigenfunctions $v_i(s)$ are an othonormal basis for $L^2(\mathcal{S})$, we can write $\boldsymbol{\phi}(s) = \sum_i \mathbf{u}_i v_i(s)$, where the Hilbert space vector $\mathbf{u}_i = \mathbb{E}_s\left[\boldsymbol{\phi}(s) v_i(s)\right]$. These vectors have magnitude

$$\|\mathbf{u}_i\|^2 = \mathbb{E}_{s,s'}\left[\langle \boldsymbol{\phi}(s), \boldsymbol{\phi}(s')\rangle v_i(s) v_i(s')\right] = \mathbb{E}_{s,s'}\left[K(s,s') v_i(s) v_i(s')\right] = \lambda_i,$$

since the v_i are kernel eigenfunctions. Thus the expected derivative magnitude

$$\begin{aligned}
\mathbb{E}_s\left[|\nabla_s \boldsymbol{\phi}(s)|^2\right] &= \mathbb{E}_s\left[\sum_i \|\mathbf{u}_i\|^2 |\nabla v_i(s)|^2\right] \\
&= \mathbb{E}_s\left[\sum_i \lambda_i v_i(s) \triangle v_i(s)\right] \\
&= \mathbb{E}_s\left[\sum_i \lambda_i \left(\sum_j a_{i,j} w_j(s)\right) \triangle \left(\sum_k a_{i,k} w_k(s)\right)\right] \\
&= \mathbb{E}_s\left[\sum_{i,j,k} \lambda_i a_{i,j} a_{i,k} w_j(s) \triangle w_k(s)\right] \\
&= \mathbb{E}_s\left[\sum_{i,j,k} \lambda_i \eta_k a_{i,j} a_{i,k} w_j(s) w_k(s)\right] \\
&= \sum_{i,j,k} \lambda_i \eta_k a_{i,j} a_{i,k} \mathbb{E}_s\left[w_j(s) w_k(s)\right] \\
&= \sum_{i,j} \lambda_i \eta_j a_{i,j}^2
\end{aligned}$$

Recall that the kernel eigenvalues λ_i are a non-increasing sequence, while the Laplacian eigenvalues η_j increase asymptotically as $n^{2/d}$. Informally, we can guess that the orthonormal matrix $a_{i,j}$ minimizing this sum will be the identity, so the sum only converges if $\lambda_n \eta_n = o(n^{-1})$ and thus $\lambda_n = o(n^{-1-2/d})$.

Formally, we define $A_j = \sum_i \lambda_i a_{i,j}^2$, so $\mathbb{E}_{s,c}\left[|\nabla_s \phi(c,s)|^2\right] = \sum_j A_j \eta_j$. Next, consider any integer $n \geq 1$, and define $B_i = \sum_{j \leq n} a_{i,j}^2$. By orthonormality of $a_{i,j}$ we therefore have $B_i \leq 1$ for all i, and also $\sum_{i=1}^{\infty} B_i \leq n$. By definition,

$$\sum_{j \le n} A_j = \sum_i \lambda_i B_i$$

Thus, $\sum_{j \le n} A_j$ cannot be more than the largest possible value of $\sum_i \lambda_i b_i$ over all b_i satisfying the same constraints as B_i: $b_i \le 1$ for all i and $\sum_i b_i \le n$. Because the λ_i are non-increasing, this maximum is achieved when $b_i = 1$ for $i \le n$ and zero thereafter, and thus

$$\sum_{j \le n} A_j \le \sum_{i \le n} \lambda_i. \tag{6}$$

Write $\Lambda_n = \sum_{i>n} \lambda_i$. By orthonormality of $a_{i,j}$, $\sum_{j=1}^{\infty} A_j = \sum_{i=1}^{\infty} \lambda_i$. Therefore (6) means that $\sum_{j>n} A_n \ge \Lambda_n$, and so

$$\eta_n \Lambda_n \le \mathbb{E}_{c,s}\left[|\nabla_s \phi(c,s)|^2\right] - \sum_{j \le n} \eta_j A_j$$

Because $\sum_{j=1}^{\infty} \eta_j A_j$ converges to $\mathbb{E}_{c,s}\left[|\nabla_s \phi(c,s)|^2\right]$, if this limit is finite then $\lim_{n\to\infty} \eta_n \Lambda_n = 0$. Because $\eta_n \sim n^{2/d}$, this means that $\Lambda_n = o(n^{-2/d})$. The same argument used at the end of theorem 4 now tells us that $\lambda_n = o(n^{-1-2/d})$. □

Note that the eigenvalue decay $\lambda_n = o(n^{-1-2/d})$ that this theorem tells us is required for finite derivative, is faster than the decay theorem 4 tells us is required for Minkowski dimension d (for all $d' > d$, $\lambda_n = O(n^{-1-2/d'})$). The following example illustrates both theorems.

Example 2. We now consider a simple example, in which a 1-dimensional circle is continuously mapped into Hilbert space, with a eigenspectrum $n^{-\alpha}$ by construction. This will show the bounds of theorem 4 cannot be improved, in the sense that for any $d \ge 1$, there is a probability distribution supported on an $\mathcal{F} \subset \mathbb{H}$ with $\overline{\dim}_M(\mathcal{F}) = d$ and covariance eigenvalues that decay $\sim n^{-1-2/d}$. This example is illustrated in Fig. 4d-f.

Parametrize the circle by an angle s, uniformly distributed between 0 and 2π. We define a vector $\boldsymbol{\phi}(s)$ in an infinite-dimensional Hilbert space, such that the covariance eigenvalues follow a power law $n^{-\alpha}$. Specifically, for $n \ge 1$:

$$\boldsymbol{\phi}(s)_{2n-1} = \frac{\cos(ns)}{n^{\alpha/2}}$$
$$\boldsymbol{\phi}(s)_{2n} = \frac{\sin(ns)}{n^{\alpha/2}}$$

The dimensions of $\boldsymbol{\phi}$ are uncorrelated, and computing their variances shows that $\lambda_{2n-1} = \lambda_{2n} = n^{-\alpha}/2$. The sine and cosine functions in each coordinate of $\boldsymbol{\phi}$ are the Laplacian eigenfunctions: the eigenfunction $\cos(ns)$ satisfies $\triangle \cos(ns) = n^2 \cos(ns)$.

Observe that $\|\boldsymbol{\phi}(s)\|^2 = \zeta(\alpha)$, where $\zeta(\alpha) = \sum_{n\ge1} n^{-\alpha}$ is the Riemann zeta function. Thus, consistent with theorem 3, population activity only has finite expectation if $\alpha > 1$. The derivative of the population code, $\left\|\frac{d\boldsymbol{\phi}(s)}{ds}\right\|^2 = \sum_{n\ge1} n^{-(\alpha-2)}$. Thus, consistent with theorem 5, the derivative only has finite expectation if $\alpha > 3$.

We now compute $\overline{\dim}_M(\mathcal{F})$. First we compute the distance between two points in $\mathcal{F}$. By circular symmetry we may without loss of generality set one of them to 0. We obtain

$$\begin{aligned} D(s)^2 = \|\boldsymbol{\phi}(s) - \boldsymbol{\phi}(0)\|^2 &= \sum_{n\ge1} \frac{(\cos(ns)-1)^2 + \sin^2(ns)}{n^\alpha} \\ &= \sum_{n\ge1} \frac{2-2\cos(ns)}{n^\alpha} \\ &= 2\zeta(\alpha) - \mathrm{Li}_\alpha(e^{is}) - \mathrm{Li}_\alpha(e^{-is}) \end{aligned}$$

where $\mathrm{Li}_\alpha(z) = \sum_{n\ge1} z^n n^{-\alpha}$ is the polylogarithm function.

To compute $\overline{\dim}_M(\mathcal{F})$ we consider how $D(s)^2$ depends on s as $s \to 0$. For non-integer α, the polylogarithm has a series expansion [28]:

$$\mathrm{Li}_\alpha(e^x) = \Gamma(1-\alpha)(-x)^{\alpha-1} + \sum_{k\geq 0} \frac{\zeta(\alpha-k)}{k!} x^k.$$

Substituting this in, we obtain

$$\begin{aligned} D(s)^2 &= -\Gamma(1-\alpha)s^{\alpha-1}(i^{\alpha-1} + (-i)^{\alpha-1}) + \zeta(\alpha-2)s^2 + O(s^4) \\ &= -2\Gamma(1-\alpha)s^{\alpha-1}\cos((\alpha-1)\pi/2) + \zeta(\alpha-2)s^2 + O(s^4) \end{aligned}$$

For any value of s we can cover $\mathcal{F}$ with $2\pi/s$ balls no smaller than $D(s)$. Thus, we have upper Minkowski dimension of d if $D(s) \sim s^{1/d}$ as $s \to 0$. For $\alpha > 3$, the dominant term as $s \to 0$ is $\zeta(\alpha-2)s^2$, so $D(s) \sim s$, and we have $\overline{\dim}_M(\mathcal{F}) = 1$. However, for $1 < \alpha < 3$, the first term of order $s^{\alpha-1}$ dominates, so $\overline{\dim}_M(\mathcal{F}) = \frac{2}{\alpha-1}$, which is greater than 1 indicating a fractal structure. At the critical value of $\alpha = 3$, we can use a Laurent expansion of the the ζ and Γ functions to obtain $\Delta(s) \sim s^2 \log \frac{1}{s}$. Thus for $\alpha = 3$, the dimension is still 1, even though $\boldsymbol{\phi}(s)$ is not differentiable.

Recall that theorem 4 predicts that for dimension d, we must have $\alpha \geq 1 + 2/d$. This is consistent with the above calculations: $d = 1$ when $\alpha \geq 3$, and $d = 2/(\alpha-1)$ when $1 < \alpha < 3$. Thus the bound is saturated for $1 < \alpha \leq 3$, but loose for $\alpha > 3$.

Example 3. In the previous example we did not explicitly model the activity of individual neurons: instead, we simply constructed a Hilbert space corresponding to their principal components, to demonstrate how differentiability and fractal dimension relate to power-law eigenspectrum decay. We now consider an example explicitly constructed from the activity of a modeled neural population with Gaussian receptive fields; this "radial basis kernel" representation is also often used in machine learning [19, 21]. We show that if all cells' receptive fields have the same size, the eigenspectrum is not a power law; however, by mixing together neurons with different radii in a scale-free manner, we obtain power-law decay, with differentiability for $\alpha > 1 + 2/d$.

Let the stimulus $\mathbf{s}$ be a point in a d-dimensional vector space, and let cell c respond according to a spherical Gaussian function, with center $\mathbf{x}_c$ and variance W. Thus the cell's response is

$$\phi(c, \mathbf{s}) = \frac{A_W}{(2\pi W)^{d/2}} e^{-|\mathbf{x}_c - \mathbf{s}|^2/2W} = A_W N(\mathbf{s}; \mathbf{x}_s, W),$$

where $N(\mathbf{x}; \mu, \sigma^2)$ represents a spherical multivariate Gaussian density of mean μ and variance σ^2, and A_W is an amplitude scaling factor that can potentially depend on the response width W. Note that this is not a model of image processing specifically: the Gaussians are not linear filters applied to an 2d image, but radial basis function units working on an abstract representation. Thus, for natural image stimuli, the stimulus space could have dimension $d \gg 2$, for example representing the results of preprocessing by a convolutional network, and the radial basis function units would then produce nonlinear receptive fields.

We will pick A_W to ensure that a cell's mean p^{th}-power activity is independent of scale: $\mathbb{E}_s\left[\phi(c,s)^p\right] = 1$ for all cells c, where p is a parameter of the model. If $p = 1$, this corresponds to the standard Gaussian normalization; while if $p = 2$, the cells' mean square firing is independent of scale. A straightforward calculation then shows that

$$A_W = V^{\frac{1}{p}} (2\pi W)^{\frac{d}{2}(1-\frac{1}{p})} p^{\frac{d}{2p}}, \tag{7}$$

where we have assumed that the stimuli are evenly distributed over a volume V. Observe that if $p = 1$, a cell's peak firing rate is larger the more sharply it is tuned (in order to obtain constant mean rate); if $p > 1$ then peak rate grows more slowly with tuning sharpness, and if $p = \infty$ then peak activity is independent of width.

Now let us compute the kernel function for two stimuli $\mathbf{s}_1$ and $\mathbf{s}_2$. We assume that the cell centers x_c are evenly distributed over the volume V, and for now consider all cells to have a fixed width W. Then

$$\begin{aligned} K_W(\mathbf{s}_1, \mathbf{s}_2) = \mathbb{E}_c\left[\phi(c, \mathbf{s}_1), \phi(c, \mathbf{s}_2)\right] &= A_W^2 V^{-1} \int N(\mathbf{s}_1; \mathbf{x}_c, W) N(\mathbf{s}_2; \mathbf{x}_c, W) d^d\mathbf{x}_c \\ &= A_W^2 V^{-1} N(\mathbf{s}_1; \mathbf{s}_2, 2W), \end{aligned}$$

where we have used the fact that the convolution of two Gaussian densities is a Gaussian of summed variance.

Now, the eigenspectrum of a translation-invariant kernel can be found simply by Fourier transformation [19, 21]. Recalling that the Fourier transform of $N(\mathbf{x}; 0, \sigma^2)$ at spatial frequency $\mathbf{k}$ is $e^{-\sigma^2|\mathbf{k}|^2/2}$, we have

$$\hat{K}_W(\mathbf{k}) = A_W^2 V^{-1} e^{-W|\mathbf{k}|^2}.$$

Plugging the distribution of spatial frequencies $\mathbf{k}$ into this function yields the eigenspectrum of the kernel function. Note that the calculation we just performed appears to give a continuous eigenspectrum. However, this is because of the "physicists'" approximation we made earlier, assuming that the volume occupied by potential stimuli was large but not explicitly modeling a probability distribution for the stimuli $\mathbf{x}_c$; had we done so, we could have obtained a discrete spectrum, but at the cost of much more work. Continuing in this informal vein, we note that the number of possible spatial frequencies of magnitude at most $|\mathbf{k}|$ will scale as $g|\mathbf{k}|^d$, where g is a geometric scale factor related to the shape and size of the distribution of $\mathbf{x}_c$. Thus, the n^{th} eigenfunction has spatial frequency magnitude $|\mathbf{k}| = (n/g)^{1/d}$. We therefore find that the eigenspectrum is

$$\lambda_n \propto \exp\left(-W(n/g)^{2/d}\right).$$

This is not a power law. For $n \ll gW^{d/2}$ it is approximately constant (as can be seen by Taylor expansion of the exponential), but for $n > gW^{d/2}$ it falls faster than any power. Thus, the eigenspectrum of a population code constructed from Gaussian neurons of a single width can be approximated by two regimes. At long distance scales, the eigenspectrum is flat, indicating a completely different population response for any two stimuli separated substantially more than the tuning width. At short distance scales however the eigenspectrum decays rapidly, indicating that the population code is barely sensitive to distances substantially shorter than the tuning width.

We may generate a power-law eigenspectrum by now considering a mixture of radial units with different variances. We model the distribution of response widths as following a power law density:

$$f(W) = \begin{cases} BW^{\gamma-1}, & \text{if } W_0 \leq W \leq W_1 \\ 0, & \text{otherwise,} \end{cases}$$

where the normalization factor $B = \gamma / \left(W_1^\gamma - W_0^\gamma\right)$. The parameter γ measures the relative distribution of sharply and broadly tuned neurons in the population: values of $\gamma > 1$ indicate a bias towards neurons of large radii, and $\gamma < 0$ indicate a bias towards small radii. We will see that the allowed range of radii $W_0 \dots W_1$ is not important provided W_0 is small enough and W_1 large enough.

We can compute the kernel corresponding to this scaled distribution as

$$\begin{aligned} \hat{K}(\mathbf{k}) = \mathbb{E}_W\left[\hat{K}_W(\mathbf{k})\right] &= \int_{W_0}^{W_1} A_W^2 V^{-1} e^{-W|\mathbf{k}|^2} BW^{\gamma-1} dW \\ &= C \int_{W_0}^{W_1} e^{-W|\mathbf{k}|^2} W^{d(1-1/p)+\gamma-1} dW, \end{aligned}$$

where the constant C captures all factors with no dependence on W or $|\mathbf{k}|$. Making a substitution $z = |\mathbf{k}|^2\, W$, we have

$$\hat{K}(\mathbf{k}) = C'\, |\mathbf{k}|^{-2d(1-1/p)-2\gamma} \int_{z_0}^{z_1} e^{-z} z^{d(1-1/p)+\gamma-1} dz. \tag{8}$$

Now, if $d(1-1/p) + \gamma > 0$, the above integral will not depend on the exact values of its limits $z_0 = |\mathbf{k}|^2\, W_0$ and $z_1 = |\mathbf{k}|^2\, W_1$: provided they are small and large enough, the integral will converge to a gamma function, $\Gamma(d(1-1/p)+\gamma)$, which shows no dependence on $|\mathbf{k}|$. We thus have $\hat{K}(\mathbf{k}) \propto |\mathbf{k}|^{-2d(1-1/p)-2\gamma}$, and

$$\lambda_n \propto n^{-\alpha},$$

where

$$\alpha = 2 - \frac{2}{p} + \frac{2\gamma}{d}. \tag{9}$$

Thus, if the basis function radii are distributed according to a scale-free power law, we obtain a power law eigenspectrum also.

To check that $\alpha > 1 + 2/d$ corresponds to finite expected derivative, we may differentiate ϕ explicitly. We find that

$$\mathbb{E}_s \left[\left| \frac{\partial \phi(c, \mathbf{s})}{\partial \mathbf{s}}^2 \right| \right] = \mathbb{E}_W \left[\frac{A_W^2}{(2\pi W)^d W^2} \mathbb{E}_{\mathbf{s}} \left[|\mathbf{x}_c - \mathbf{s}|^2 e^{-|\mathbf{x}_c - \mathbf{s}|^2/W} \right] \right]$$

Because $\mathbb{E}_{\mathbf{s}} \left[|\mathbf{x}_c - \mathbf{s}|^2 e^{-|\mathbf{x}_c - \mathbf{s}|^2/W} \right] \propto W^{1+d/2}$, and using (7),

$$\begin{aligned} \mathbb{E}_s \left[\left| \frac{\partial \phi(c, \mathbf{s})}{\partial \mathbf{s}}^2 \right| \right] &\propto \mathbb{E}_W \left[W^{d(1/2-1/p)-1} \right] \\ &\propto \int_{W_0}^{W_1} W^{\gamma + d(1/2-1/p)-2} \, dW, \end{aligned}$$

This integral will diverge as $W_0 \to 0$ unless $\gamma > 1 - d(1/2 - 1/p)$, which from (9) we see is equivalent to $\alpha > 1 + 2/d$. Thus, confirming theorem 5, the expected derivative will be infinite if eigenvalues do not decay faster than $n^{-1-2/d}$, and this infinite derivative will reflect unbounded contributions from the neurons with the smallest values of W, i.e. neurons of sharpest tuning.

3. The $N \to \infty$ limit

To make formal statements about the asymptotic decay of the eigenspectrum requires defining a limit in which both the number of neurons recorded, and the number of stimuli presented grow large. While most statistical analyses involve a limit of a large number of observations, the fact that we consider two such limits means we must consider not just randomly distributed numbers, but randomly distributed vectors in infinite-dimensional spaces. This approach, known as functional data analysis, has not yet been commonly applied in neuroscience, but has a well-developed theoretical basis thanks to applications in other domains [29, 4].

The logic underlying functional data analysis is the standard logic of statistical inference: using the properties of a limited sample to make inferences about a larger, unobserved population. Specifically, by analyzing how the particular neurons we recorded responded to the particular stimuli we presented, we infer facts about how a larger population of neurons would respond to a larger ensemble of stimuli. To do so, we employ a standard assumption: that our sample of neurons and stimuli are randomly sampled from the larger population and the larger ensemble.

A simple example of inference by random sampling would be estimating the distribution of lengths our mice's tails. If we were to say that these lengths followed a Gaussian distribution (or indeed any continuous distribution), that could not be correct: we studied a finite number of mice, so their tail lengths follow a discrete distribution. However, we can use the sample we measured to make an inference about the population they are drawn from. This larger population is not just the set of all mice in the lab colony, or even all mice currently alive, both of which are finite. It is a hypothetical infinite population of all mice that could have been tested – only such an infinite population can follow a continuous probability distribution such as a Gaussian.

The assumption of such an infinite population can have counter-intuitive consequences, some of which are relevant for our theorems. For example, distributions of wealth and income can be modeled by a Pareto distribution[30]: $\mathbb{P}(x) \propto x^{-1-\gamma}$. When $\gamma \leq 2$, this distribution has infinite variance, $\mathbb{E}\left[x^2\right] = \infty$. Any finite sample drawn from this distribution will have finite variance, but the variance of the sample will continue increasing without bound as the sample size grows. Conversely, a statistician who found that sample variance converges to a finite limit as sample size grows could infer that the population has finite variance.

In our case, we consider the visual stimuli presented to be drawn at random from a probability distribution. We would like to make an inference about neural coding of not just these stimuli, nor even of the entire ImageNet database, but rather of a hypothetical infinite population of similar natural images we could in principle have presented. If we find that properties of the observed neural code converge to a limit as the number of images analyzed increases, we infer that it can be modelled by a probability distribution predicting this limit.

We similarly consider the recorded neurons to be drawn randomly from a population distribution. Again, this population is not just the full set of V1 neurons in the particular mouse studied, but the infinite set of neurons which that mouse's brain might have contained, consistent with our experimental sample. Again, if we find that properties of the observed neural code converge to a limit as the number of neurons analyzed increases, we infer that this infinite population can be modelled by a probability distribution predicting this limit.

The setting for our theorems thus involves three elements. First, a probability distribution for possible stimuli $\mathbb{P}(s)$, over a hypothetically infinite stimulus set $\mathcal{S}$; second, a probability distribution for possible recorded cells $\mathbb{P}(c)$, over a hypothetically infinite cell set $\mathcal{C}$; and third, a function $\phi(c, s)$ that predicts the expected firing rate of cell c to stimulus s. (As before, $\phi(c, s)$ describes the neuron's deterministic mean response to the stimulus, excluding neuronal noise). It is of course impossible to obtain a complete description of these three elements, which would summarize the response of any conceivable V1 cell to any conceivable natural image stimulus. However, it is possible to infer some of their properties using the logic of random sampling described above.

3.1 Fully independent coding revisited

The random sampling framework allows us to mathematically formalize the scenario of example 1, showing that if a set of neurons independently code a single continuous variable, the kernel function becomes discontinuous in a limit that the number of neurons tends to infinity. Constructing this limit requires some mathematical subtleties. If we considered the neurons as sampled with replacement from a finite set of N possibilities, the eigenspectrum would not be flat: once the sample becomes larger than N it would contain multiple cells whose activity would be perfectly correlated. The eigenspectrum would thus be flat for the first N eigenvalues, and exactly zero thereafter. Similarly, if we considered the neurons to be drawn from a discrete and (countably) infinite distribution, a large enough population would contain neurons with identical tuning properties, and thus all neurons could not be independent. This means that the binary coding scheme described in example 1 cannot be taken to a limit of infinite neurons.

Example 4. We therefore consider neurons as drawn from a continuous probability distribution. Again with the stimulus s uniformly distributed between 0 and 1, let each cell c respond to each possible stimulus s with an independent standard Gaussian variate. Cell c's tuning function $\phi(c, s)$ is then with probability 1 discontinuous, and not even measurable (a set theoretic requirement that allows it to be integrated). The space of possible neural responses is thus not a subset of the Hilbert space $L^2(\mathcal{S})$, but rather of a much larger space $\mathcal{C} = \mathbb{R}^{[0,1]}$ of all set-theoretic functions from the interval $[0, 1]$ to the real numbers. The kernel function is $K(s, s') = 1_{s=s'}$, the function that is 1 if $s = s'$, and 0 otherwise, reflecting the fact that neurons have variance 1, but that population responses to two unequal stimuli are completely uncorrelated, however similar these stimuli are. This kernel function is discontinuous, and the code is completely unable to generalize; a response learned to the representation of one stimulus has absolutely no bearing on the response to other stimuli.

The eigenvalues of the kernel function are all zero, since the function $1_{s_1=s_2}$ is zero everywhere except a set of measure 0. Thus, $\sum_{i=1}^{\infty} \lambda_i \neq \mathbb{E}_{s,c}\left[\phi(c, s)^2\right]$, but theorem 3 is not violated since the requirement on kernel continuity was not satisfied. To understand how the kernel eigenvalues can be zero despite unit population variance, consider the eigenvalues of the sample kernel matrix $K_{i,j} = \mathbb{E}_c\left[\phi(c, s_i)\phi(c, s_j)\right]$. This is an identity matrix, so the eigenvector equation $\frac{1}{N_s}\sum_{j=1}^{N_s} K_{i,j} v_{n,j} = \lambda_n v_{n,i}$ implies that all eigenvalues $\lambda_n = \frac{1}{N_s}$, and thus indeed tend to zero as $N_s \to \infty$. The fact that the eigenvalues approach zero reflects the fact that none of each cell's variance is shared with the rest of the population. A discussion of such representations from the perspective of machine learning can be found in Ref. [31].

Remark. The condition on continuity of K in theorem 3 can actually be relaxed [32]. However, the eigenvalue sum is then equal to the diagonal expectation of a "smoothed" version of the kernel function, that essentially removes points of discontinuity. The smoothed version of $1_{s=s'}$ will be identically zero, so again has only zero eigenvalues.

3.2 Infinite population variance

Theorem 3 describes two conditions that can lead to eigenspectra decaying slower than n^{-1}: a discontinuous kernel function, or infinite population variance. We now consider the second possibility.

As with the Pareto distribution example raised earlier, infinite population variance does not require any neuron to fire at an infinite rate. However it does imply that there must be cells responding to their preferred stimuli at arbitrarily large rates. Although such cells would be increasingly rare as the peak rate become larger, their activity is strong enough to dominate the expected variance of the population. This possibility of course could not actually happen due to physical constraints on neuronal firing, but it is still instructive to consider a counterfactual example. In this example, an infinite set of possible stimuli are represented by dedicated neuronal populations of increasing sparseness. We show that slowly decaying eigenvalues can be obtained if neurons responding to increasingly rarer stimuli do so with increasingly large firing rates. By varying the parameters of the model, we illustrate how eigenspectra decaying as n^{-1} or slower require this increase to be so rapid that population variance diverges to an infinite sum.

Example 5. We consider a discrete, infinite distribution of possible stimuli, each labeled by an positive integer. The stimuli are presented at random, with the probability that stimulus s is presented on any trial following a *zeta* *distribution*:

$$\mathbb{P}(s) = \frac{s^{-\alpha}}{\zeta(\alpha)}.$$

The zeta distribution gets its name from the normalizing factor $\zeta(\alpha) = \sum_{n=1}^{\infty} n^{-\alpha}$, which is known as the Riemann zeta function. The zeta distribution is a discrete power law: the probability of presenting stimulus s is proportional to $s^{-\alpha}$. The distribution is only defined if $\alpha > 1$. It is a "heavy tailed" distribution: while stimulus $s = 1$ is always the most likely to be presented, when $\alpha \approx 1$ the distribution of stimuli will be very wide compared to common probability distributions such as the Gaussian or Poisson, whose probabilities decay as an exponential rather than power law. The zeta distribution has infinite mean if $\alpha \leq 2$, and infinite variance if $\alpha \leq 3$; more generally, for any value of α, all moments above $\alpha - 1$ will be infinite.

We consider each stimulus to be represented by a dedicated set of cells, which respond only to that stimulus. These cell sets are therefore also labelled with positive integers. We assume that cells are also drawn according to a zeta distribution, but this time with a different parameter β. Thus, the fraction of cells in the population belonging to cell class c is:

$$\mathbb{P}(c) = \frac{c^{-\beta}}{\zeta(\beta)}.$$

Finally, the response of a cell of class c when stimulus s is presented is modeled as

$$\phi(c, s) = s^{\gamma/2}\delta_{s,c},$$

so cells of class c respond when $s = c$, but not to any other stimuli. Cells with large values of c respond only to rare stimuli, but if $\gamma > 0$, they respond especially strongly to them. The kernel function is

$$K(s, s') = \mathbb{E}_c\left[\phi(c, s)\phi(c, s')\right] = \frac{s^{\gamma-\beta}\delta_{s,s'}}{\zeta(\beta)}.$$

Its eigenfunctions are $v_n(s) = \sqrt{n^{\alpha}\zeta(\alpha)}\delta_{s,n}$, which are easily shown to be orthonormal: $\mathbb{E}_s\left[v_n(s)v_m(s)\right] = \delta_{n,m}$, with eigenvalues

$$\lambda_n = \frac{n^{\gamma-\alpha-\beta}}{\zeta(\alpha)\zeta(\beta)}.$$

The expected squared population activity is

$$\begin{aligned} \mathbb{E}_{c,s}\left[\phi(c,s)^2\right] &= \sum_{c,s} \frac{s^{-\alpha}c^{-\beta}}{\zeta(\alpha)\zeta(\beta)} s^{\gamma}\delta_{c,s} = \sum_{s} \frac{s^{\gamma-\alpha-\beta}}{\zeta(\alpha)\zeta(\beta)} \\ &= \frac{\zeta(\alpha+\beta-\gamma)}{\zeta(\alpha)\zeta(\beta)} = \sum_n \lambda_n, \end{aligned} \tag{10}$$

so the eigenspectrum sum equals the expected squared activity, as predicted by theorem 3. (Note that even though the stimulus space is discrete, from a mathematical perspective the requirement of continuous kernel function still holds, as the stimulus space has the *discrete topology*[22].)

The eigenvalues decay as $n^{\gamma-\alpha-\beta}$, and thus will have an infinite sum if and only if $\gamma \geq \alpha + \beta - 1$. Recall that γ determines the extent to which rarer stimuli produce stronger responses, while α and β determine how rare are these stimuli and the cells responding to them. Infinite eigenvalue sum therefore occurs only if these sparsely-firing neurons respond so strongly to their preferred stimuli that they still dominate average population activity.

3.3 Holographic coding for finite-sum eigenspectra

Examples 1 and 5 illustrate that the pathology of neural codes with infinite eigenvalue sums becomes more apparent as the population size gets larger. In example 1, we see this directly: the larger the population of independent cells, the larger the fraction of cells devoted to representing irrelevant stimulus details. With the infinite variance of example 5, the implications of large population size require more careful analysis. As with the Pareto model, the activity variance of a random sample of cells would continue to increase without limit as the sample size grows. Notwithstanding physical limitations on neural firing rates, if an actual neural code had this property then accurately gauging the population's response to stimuli would require sampling the arbitrarily small fraction of cells giving the largest responses. Such a code would be too sparse to be of any use: unless we (or a downstream brain structure) could sample the entire population, we will always miss the most important cells.

Theorem 6 formalizes this concept. It shows that in a neural code with finite population variance, the activity of a sufficiently large random sample of cells suffices to predict the activity of the entire population to arbitrary expected accuracy. We term this the "holographic" property, analogously to how a small fraction of an optical hologram can reconstruct a full image. We conclude that finite-variance codes are suited for structures such as cortex, where downstream neurons can sample only a subset of the population's activity. Conversely, codes not satisfying the n^{-1} bound are pathological in that they require downstream structures to sample every neuron to read out the population code.Examples 4 and 5 illustrate how this fails for codes whose eigenspectra decay too slowly: in the first case, where the eigenspectrum is flat, all neurons are independent so no one can be predicted from any other; in the second case, however many neurons one samples from, there will always be more neurons responding arbitrarily strongly to rare stimuli that did not drive any cells in the training sample.

Theorem 6. *Let $\phi(c, n)$ be a continuous code with finite eigenvalue sum, let $\{c_i\}$ be a sequence of cells drawn independently at random from the cell population $\mathcal{C}$, and let $\boldsymbol{\phi}_n(s)$ be the vector containing the responses of the first n cells to stimulus s. With probability 1, there exists for each possible target cell c a sequence of weight vectors $\mathbf{w}_n(c)$ such that*

$$\lim_{n\to\infty} \mathbb{E}_{c,s}\left[|\mathbf{w}_n(c)\cdot\boldsymbol{\phi}_n(s) - \phi(c,s)|^2\right] = 0.$$

Conversely, if the eigenvalue sum is infinite, no such sequence $\mathbf{w}_n(c)$ exists.

Proof. First note that with probability 1 over c, $\phi(c,\cdot) \in L^2(\mathcal{S})$. Indeed, if this were not the case then $\mathbb{E}_{c,s}\left[\phi(c,s)^2\right]$ would be infinite, contradicting finite eigenvalue sum.

Now for each cell c, define the weight vector $\mathbf{w}_n(c)$ by standard multivariate linear regression, as the weights minimizing $\mathbb{E}_s\left[|\hat{\phi}_n(c,s) - \phi(c,s)|^2\right]$, where $\hat{\phi}_n(c,s) = \mathbf{w}_n(c)\cdot\boldsymbol{\phi}_n(s)$. As usual for linear regression, the prediction $\hat{\phi}(c,s)$ is a projection of the target variable onto the space spanned by the predictor variables: $\hat{\phi}_n(c,\cdot) = \mathscr{P}_n\phi(c,\cdot)$, where $\mathscr{P}_n$ represents the projection operator onto the subspace of $L^2(\mathcal{S})$ spanned by the functions $\phi(c_1,\cdot),\ldots,\phi(c_n,\cdot)$.

The expected error over all possible target cells and stimuli is thus

$$\begin{aligned}\mathbb{E}_{c,s}\left[|\hat{\phi}_n(c,s) - \phi(c,s)|^2\right] &= \mathbb{E}_c\left[\|\phi(c,\cdot) - \mathscr{P}_n\phi(c,\cdot)\|^2\right] \\ &= \mathbb{E}_c\left[\|(\mathscr{I} - \mathscr{P}_n)\phi(c,\cdot)\|^2\right] \\ &= \mathrm{Tr}\big(\mathscr{K}(\mathscr{I} - \mathscr{P}_n)\big), \end{aligned} \tag{11}$$

where $\mathscr{I}$ represents the identity operator, $\mathscr{K}$ the kernel operator, and $\|\cdot\|$ the Hilbert space norm of $L^2(\mathcal{S})$. Now, define the sample kernel function $K_n(s,s') = \frac{1}{n}\sum_{i=1}^n \phi(c_i,s)\phi(c_i,s')$, and let $\mathscr{K}_n$ be the corresponding integral operator. $\mathscr{K}_n$ has rank n, and projection onto the space spanned by its non-zero eigenfunctions is equivalent to application of the projection operator $\mathscr{P}_n$. By theorems 8.1.2 and 5.1.4 of Ref. [4], we can therefore conclude that the operator $\mathscr{I} - \mathscr{P}_n$ converges to zero on the range of $\mathscr{K}$, but only in a particular sense. It does not converge in *operator norm*, which would mean that for any $\epsilon > 0$, there exists an N such that $\|(\mathscr{I} - \mathscr{P}_n)x\| < \epsilon\|x\|$ for all $n \geq N$ and $x \in \mathrm{range}(\mathscr{K})$; indeed, $\mathscr{P}_n$ is a finite rank operator, so if $\mathscr{K}$ is infinite rank there is always an $x \in \mathrm{range}(\mathscr{K})$ with $(\mathscr{I} - \mathscr{P}_n)x = x$. Instead, for any m and ϵ, there exists an N such that $\|(\mathscr{I} - \mathscr{P}_n)x\| < \epsilon\|x\|$ for all $n \geq N$ and for all x in the space spanned by $v_1,\ldots,v_m$, the first m eigenfunctions of $\mathscr{K}$.

If the eigenvalues of $\mathscr{K}$ have finite sum, this mode of convergence is sufficient for the expected prediction error (11) to converge to zero. Indeed, to obtain error less than ϵ, choose m so that $\sum_{i>m}\lambda_i \leq \epsilon/2$, and choose N so that

$\|(\mathscr{I}-\mathscr{P}_n)x\| \leq \delta \|x\|$ for all $n \geq N$ and all x in the space spanned by $v_1, \ldots, v_m$, where $\delta = \frac{\epsilon}{2\sum_{i\leq m}\lambda_i}$. As the v_i form an orthonormal basis for $L^2(\mathcal{S})$,

$$\begin{aligned}
\mathrm{Tr}\big((\mathscr{I}-\mathscr{P}_n)\mathscr{K}\big) &= \sum_{i=1}^{\infty}\langle v_i, (\mathscr{I}-\mathscr{P}_n)\mathscr{K} v_i\rangle \\
&= \sum_{i\leq m}\lambda_i\langle v_i, (\mathscr{I}-\mathscr{P}_n)v_i\rangle + \sum_{i>m}\lambda_i\langle v_i, (\mathscr{I}-\mathscr{P}_n)v_i\rangle \\
&\leq \sum_{i\leq m}\lambda_i\delta + \sum_{i>m}\lambda_i \\
&\leq \epsilon/2 + \epsilon/2 = \epsilon.
\end{aligned}$$

Conversely, if the eigenvalues of $\mathscr{K}$ have infinite sum, then because $\mathscr{P}_n$ is a projection of rank n, the Courant-Fischer theorem (Ref. [4], theorem 4.2.7) tells us that

$$\mathrm{Tr}\big((\mathscr{I}-\mathscr{P}_n)\mathscr{K}\big) \geq \sum_{i>n}\lambda_i = \infty.$$

Therefore, however many neurons n we sample from, our expected prediction error is unlimited. □

3.4 Pathologies of infinite derivative codes

Similar arguments to those of sections 3.2 and 3.3 illustrate the pathologies of codes with eigenspectra decaying slower than $n^{-1-2/d}$, which have infinite expected deriviate magnitude (theorem 5). To have an infinite population derivative does not require the tuning curve of any individual neuron to be nonsmooth. Indeed, as we saw in example 3, it is possible for each neuron's stimulus responses to be differentiable, but the population code not to be. In this scenario, the population contains neurons of ever sharper tuning, so though the total derivative is finite for any finite sample of neurons, it continues to grow without bound as the sample size increases. Fractal population codes are therefore further pathological for reasons analogous to theorem 6, and as illustrated in example 3: the representation of the derivative is dominated by increasingly rare neurons of arbitrarily sharp tuning. Reliably reading out the difference in the representations of two similar stimuli therefore requires sampling essentially the entire population, or these rare cells will be missed.

To see why non-differentiability would be pathological for read-out of a neural code, consider the problem of trying to distinguish two nearby stimuli s and s' which differ by an amount ds. The difference in their firing patterns will be approximately $\frac{\partial\phi(c,s)}{\partial s}ds$, and for a downstream brain to distinguish the two stimuli, it must be able to specifically weight those cells for which $\frac{\partial\phi(c,s)}{\partial s}$ is large. However, if $\mathbb{E}_c\left[\left|\frac{\partial\phi(c,s)}{\partial s_j}\right|^2\right] = \infty$, the neural representation of the derivative is dominated by a sparse set of cells of arbitrarily high derivative: for any D, there are cells for which $\left|\frac{\partial\phi(c,s)}{\partial s_j}\right|^2 > D$, and these cells dominate the total activity $\mathbb{E}_c\left[\left|\frac{\partial\phi(c,s)}{\partial s_j}\right|^2\right]$. The difference in population responses to s and s' gets smaller the closer s' is to s, however this difference becomes concentrated in an ever sparser subset of cells. Such a code is pathological, for the same reason as for infinite variance codes: a downstream structure will not be able to accurately respond to the difference between s' and s without sampling all the cells in the population.

3.5 Implications for coding in finite populations

The discussion so far has focused on the limit of how the eigenspectrum λ_n scales as $n \to \infty$. However, we can only present a finite number of stimuli, and the brain only contains a finite number of neurons. This section focuses on interpretation of these results for coding by finite populations.

Our experimental results (Figs. 2g-j, Extended Data Fig. 10, Extended Data Fig. 8) show that the power law holds accurately over a progressively larger range the more stimuli are analyzed and the more neurons are recorded; typically this range extends to approximately half the total number of stimuli presented. Theorem 2 suggests that the the reason eigenvalues fall below the power law at this point is because the remaining eigenvectors correspond to or are corrupted by additive noise, and thus will have cross-validated eigenvalues approaching zero.

Because the number of stimuli we presented was determined by the (scientifically arbitrary) maximum time we could perform our experiments, it is reasonable to infer that the powerlaw would hold over a larger range if more stimuli had been presented. Eventually, however, the powerlaw must stop: the mouse's brain contains only a finite number of neurons, and the number of stimuli presentable on the monitor is finite. We also cannot rule out the possibility that, if we were able to present sufficiently many stimuli, we would observe the eigenspectrum would at some point reliably deviate from a power law, in a manner that did not simply result from measurement noise.

The question of how to interpret a scaling law that holds over a finite range is familiar from physics. Scaling laws abound in statistical mechanics, but almost always over a finite range due to the particulate nature of matter. Particularly relevant to the present case is the example of phase transitions. Phase transitions – abrupt changes of state occurring when a system parameter crosses a particular value – only theoretically occur in infinite systems [33]. Finite systems show continuous dependence on the parameter value, but this dependence grows sharper the larger the system is. Thus, for finite but large systems, the dependence on the parameter is smooth but so rapid as to appear discontinuous to close approximation.

The present case is analogous. Consider an eigenspectrum that decayed as $\lambda_n \propto n^{-\alpha}$ over a finite range $n = 1$ to N. The total population variance is always finite, and an integral approximation gives

$$\sum_{n=1}^{N} n^{-\alpha} \approx \int_1^N n^{-\alpha} dn = \frac{N^{1-\alpha} - 1}{1-\alpha}.$$

This is a continuous function of α, for any finite value of N (including $\alpha = 1$, where Taylor expansion yields $\log(N)$). The actual value of N is unimportant if $\alpha > 1$, as the sum will always be close to $1/(\alpha - 1)$. However, we observe a rapid increase in total variance when α becomes close to 1, and this increase will be more rapid the larger N is. In the limit $N \to \infty$, we obtain a discontinuous "phase transition" to infinite variance when $\alpha \geq 1$.

A neural code whose eigenspectrum decayed more slowly than n^{-1} over a large but finite range would still be pathological, for the same reasons as described in theorems 3 and 6 and examples 1 and 5. The great majority of population variance would come from the tail eigenfunctions, devoted to encoding tiny details of stimuli; this failure would not occur discontinuously at $\alpha \geq 1$, but its dependence on α would grow sharper the larger N is. Analogous arguments show that for a code whose eigenspectrum decayed more slowly than $n^{-1-2/d}$, the great majority of the derivative's variance would come from the tail eigenfunctions, resulting in a poor code for the reasons given in the discussion of theorems 4 and 5, section 3.4, and examples 2 and 3.

Finally, we note that these arguments apply only when the number of neurons in the population is substantially greater than the manifold dimension of the stimulus set. As described in example 1, encoding a stimulus of manifold dimension d to an accuracy of 1 in 1024 will take $10d$ neurons. If $d \ll N$ then nearly all the population variance is devoted to encoding irrelevant stimulus details; but if $d \approx N$ this need not be the case. We therefore hypothesize that the eigenspectrum bounds we observed in visual cortex are likely to hold in any brain structure where the number of neurons substantially exceeds the manifold dimension of the stimulus being encoded.

3.6 cvPCA in the limit $N_c \to \infty$

We can use the random-sampling framework to analyze the cvPCA algorithm in the limit that the number of both cells and stimuli grows large. We will extend theorems 1 and 2 to this limit, showing that the eigenspectrum estimated from a finite sample converges to the eigenspectrum of the continuous kernel function; and that unbiased estimation of the population eigenspectrum does not require all cells to have equal uncorrelated noise variance in this limit.

In the $N_c \to \infty$ limit, we can define both a kernel function $K(s, s') = \mathbb{E}_c [\phi(c, s)\phi(c, s')]$, where s and s' range over the entire population of possible stimuli, and a correlation function $G(c, c') = \mathbb{E}_s [\phi(c, s)\phi(c', s)]$ where c and c' range over the entire population of potential cells. The kernel function and correlation function have the same eigenvalues: if the kernel eigenfunctions satisfy $\mathbb{E}_{s'} [K(s, s')v_n(s')] = \lambda_n v_n(s)$, then correlation eigenfunctions $u_n(c)$ are related to the kernel eigenfunctions by $u_n(c) = \lambda_n^{1/2}\mathbb{E}_s [\phi(c, s)v(s)]$, and have the same eigenvalues. Thus $\mathbb{E}_{c'} [G(c, c')u_n(c')] = \lambda_n u_n(c)$, and $\mathbb{E}_c [u_n(c)u_m(c)] = \delta_{n,m}$. The fundamental result underpinning the current study is that the eigenspectrum of a finite kernel matrix, defined by a random sample of stimuli and neurons, converges to the eigenspectrum of the full kernel function as the size of the sample increases. This convergence is guaranteed to occur if population variance is finite, $\mathbb{E}_{c,s} \left[\phi(c, s)^2\right] < \infty$, and can also happen under less restrictive conditions [34]. Thus, by observing how a sufficiently large number of neurons respond to a sufficiently large number of stimuli, we can make inferences about the entire population code, defined by the responses of any neurons we might have recorded in that mouse, to any natural image stimuli we might have presented with similar properties.

The generalization of theorem 1 to an infinite population closely mirrors the original:

Theorem 7. *Let $\{s_i : i \in \mathbb{N}\}$ be an infinite sequence of cells drawn from $\mathbb{P}(\mathcal{S})$, and $\{c_j : j \in \mathbb{N}\}$ be an infinite sequence of cells drawn from $\mathbb{P}(\mathcal{C})$. Let $\hat{\lambda}_n$ denote the n^{th} eigenvalue estimated by cvPCA from a sample of the first N_s stimuli and N_c cells in this sequence, and let λ_n represent the n^{th} eigenvalue of the full kernel function. Then with probability 1,*

$$\lim_{N_s,N_c\to\infty} \mathbb{E}_{\nu_1,\nu_2}\left[\sum_{n=1}^{N}\hat{\lambda}_n\right] \le \sum_{n=1}^{N}\lambda_n. \tag{12}$$

Proof. To show convergence of the estimated eigenspectrum as the number of cells and stimuli grows larger, we will make use of versions of the *strong law of large numbers* (SLLN). The simplest version of the SLLN concerns a series independent random variables X_i drawn from a common distribution, and says that with probability 1 the sample means $\frac{1}{N}\sum_{i\le N} X_i$ converge to $\mathbb{E}[X]$ as $N \to \infty$, provided $\mathbb{E}[|X|]$ is finite. Note that the SLLN applies to single draws of the sequence X_i: it says that there is no chance that the sample means will not converge, for any single draw. Here, we will apply this philosophy to the stimuli and neurons drawn from the population distributions, showing that with probability 1 we expect convergence as we analyze larger sets of stimuli and neurons.

Specifically, we will make use of two specialized forms of SLLN. The first (Ref. [4], theorem 8.1.2) concerns covariance functions, and says if $\mathbb{E}_{c,s}\left[f(c, s)^2\right] < \infty$, then with probability 1, $\lim_{N_s\to\infty} \frac{1}{N_s}\sum_{i\le N_s} \mathrm{f}_1(s_i, c)\mathrm{f}_1(s_i, c')$ converges to the total covariance function $\tilde{G}(c, c')$ in Hilbert-Schmidt norm, i.e. $\mathbb{E}_{c,c'}\left[\left(\sum_{i\le N_s} \mathrm{f}_1(s_i, c)\mathrm{f}_1(s_i, c') - \tilde{G}(c, c')\right)^2\right] \to 0$. This is sufficient for its eigenfunctions to converge to the eigenfunctions $\tilde{u}_n(c)$ of the covariance function $\tilde{G}(c, c')$.

The second form of SLLN concerns the eigenvectors of the sample correlation matrices[35]. It says that under the same finite-variance assumption, the eigenvectors of the sample covariance matrix will converge with probability 1 to the population eigenfunctions, in the mean-square sense. Putting the two results together, we can conclude that

$$\lim_{N_c,N_s\to\infty} \frac{1}{N_c}\sum_{i\le N_c} (\tilde{\mathbf{u}}_{n,i} - \tilde{u}_n(c_i))^2 = 0. \tag{13}$$

Now let us consider the n^{th} cross-validated eigenvalue, $\hat{\lambda}_n = (\mathbf{F}_1\tilde{\mathbf{u}}_n) \cdot (\mathbf{F}_2\tilde{\mathbf{u}}_n)/N_s N_c^2$. Taking expectations over the noise on both repeats, we obtain

$$\mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right] = \frac{1}{N_s N_c^2}\sum_{k\le N_s}\mathbb{E}_{\nu_1,\nu_2}\left[\Big(\sum_{i\le N_c}\big(\phi(c_i, s_k) + \nu_1(c_i, s_k)\big)\tilde{\mathbf{u}}_{n,i}\Big)\Big(\sum_{j\le N_c}\big(\phi(c_j, s_k) + \nu_2(c_j, s_k)\big)\tilde{\mathbf{u}}_{n,j}\Big)\right]$$

Now, because ν_2 is independent of ν_1 and $\tilde{\mathbf{u}}_n$, it contributes nothing to the expectation. Thus,

$$\mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right] = \frac{1}{N_s N_c^2}\sum_{k\le N_s, i\le N_c, j\le N_c}\mathbb{E}_{\nu_1}\left[\phi(c_i, s_k)\phi(c_j, s_k)\tilde{\mathbf{u}}_{n,i}\tilde{\mathbf{u}}_{n,j} + \nu_1(c_i, s_k)\phi(c_j, s_k)\tilde{\mathbf{u}}_{n,i}\tilde{\mathbf{u}}_{n,j}\right]$$

Because the sample eigenvector $\tilde{\mathbf{u}}_n$ is computed from repeat 1, it is correlated with ν_1, so the expectation of the second term is not zero. Nevertheless, the size of this expectation converges to zero for large enough numbers of cells and stimuli. Indeed, equation (13) shows that $\tilde{\mathbf{u}}_{n,i}$ approaches the population eigenfunction $\tilde{u}_n(i)$ in mean-square, and $\tilde{u}_n(i)$ is independent of ν_1. Applying the Cauchy-Schwartz inequality and the fact that ν_1 has finite variance, we see that $\frac{1}{N_c}\sum_{i\leq N_c}\nu_1(c_i,s_k)(\tilde{\mathbf{u}}_{n,i}-\tilde{u}_n(i))$ tends to zero, so we may replace $\tilde{\mathbf{u}}_{n,i}$ by its limit, obtaining

$$\begin{aligned}\lim_{N_c,N_s\to\infty}\mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right] &= \lim_{N_c,N_s\to\infty}\frac{1}{N_sN_c^2}\sum_{k\leq N_s,i\leq N_c,j\leq N_c}\phi(c_i,s_k)\phi(c_j,s_k)\tilde{u}_n(c_i)\tilde{u}_n(c_j)\\ &= \mathbb{E}_{s,c,c'}\left[\phi(c,s)\phi(c',s)\tilde{u}_n(c)\tilde{u}_n(c')\right]\\ &= \mathbb{E}_{c,c'}\left[G(c,c')\tilde{u}_n(c)\tilde{u}_n(c')\right] \qquad (14)\end{aligned}$$

If the functions $\tilde{u}_n$ were eigenfunctions of the signal covariance operator $K(c,c')$, then this last term would equal the signal population eigenvalues λ_n. However, $\tilde{u}_n$ are the eigenfunctions of the total covariance operator $\tilde{K}(c,c')$. As we discuss below in Theorem 8, these are likely to coincide for situations such as the current recording. However, even if they do not coincide, we may obtain a one-sided bound. Indeed, the Courant-Fisher minimax principle (Ref. [4], theorem 4.2.7) implies that for any set of orthonormal functions $\tilde{u}_n(c)$, this expectation cannot exceed what would be obtained with the true eigenfunctions: $\sum_{n\leq N}\mathbb{E}_{c,c'}\left[K(c,c')\tilde{u}_n(c)\tilde{u}_n(c')\right]\leq\sum_{n\leq N}\lambda_n$. □

The generalization of theorem 2 closely mirrors the original, but without the requirement for identical noise variances.

Theorem 8. *Consider a noise model*

$$\nu_r(c,s)=\alpha_r(s)\phi(c,s)+\beta_r(c,s)+\gamma_r(c,s)$$

where $\alpha_r(s)$ represents multiplicative noise scaling the entire population's response to stimulus s on repeat r; $\beta_r(c,s)$ is additive noise in dimensions orthogonal to the stimulus; and $\gamma_r(c,s)$ is independent between neurons and stimuli. Assume that with α, β and γ statistically independent of each other and of $\phi(c,s)$, but let the variance of γ vary between cells. Then the eigenvalue estimates $\mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right]$ converge to the population eigenvalues λ_n together with additional zero eigenvalues corresponding to the additive noise dimensions.

Proof. From equation (14), we see that if an eigenfunction $\tilde{u}_n(c)$ of the total covariance function $\tilde{G}(c,c')$ is equal to an eigenfunction $u_m(c)$ of the signal covariance function $G(c,c')$, then $\lim_{N_s,N_c\to\infty}\mathbb{E}_{\nu_1,\nu_2}\left[\hat{\lambda}_n\right]=\lambda_m$. It therefore suffices to show that the eigenfunctions of $\tilde{G}(c,c')$ are the same as those of $G(c,c')$, together with additional orthogonal dimensions. To do so, note that the total response

$$f_r(c,s)=\left(1+\alpha_r(s)\right)\phi(c,s)+\beta_r(c,s)+\gamma_r(c,s)$$

Thus,

$$\begin{aligned}\tilde{G}(c,c') &= \mathbb{E}_{s,\alpha,\beta,\gamma}\left[f_r(c,s)f_r(c',s)\right]\\ &= \mathbb{E}_{s,\alpha}\left[(1+\alpha_r(s))^2\phi(c,s)\phi(c',s)\right]+\mathbb{E}_{s,\beta}\left[\beta_r(c,s)\beta_r(c',s)\right]+\mathbb{E}_{s,\gamma}\left[\gamma_r(c,s)\gamma_r(c',s)\right]\\ &= V_\alpha G(c,c')+B(c,c')+V_\gamma(c)1_{c=c'}.\end{aligned}$$

Here, $V_\alpha=\mathbb{E}_{\alpha,s}\left[(1+\alpha_r(s))^2\right]$ is a scale factor resulting from multiplicative modulation, $B(c,c')$ represents the correlation of the additive noise, $V_\gamma(c)=\mathbb{E}_{s,\gamma}\left[\gamma_r(c,s)^2\right]$ is the independent noise variance, and $1_{c=c'}$ represents the function that is 1 if $c=c'$, zero otherwise. Note that $\mathbb{E}_{c'}\left[B(c,c')u(c')\right]=0$ as we have assumed the additive noise dimensions orthogonal to the signal dimensions, and $\mathbb{E}_{c'}\left[1_{c=c'}u(c')\right]=0$ since $\mathbb{P}(c=c')=0$. Thus, the signal eigenfunctions $u_n(c)$ are eigenfunctions of $\tilde{K}$:

$$\mathbb{E}_{c'}\left[\tilde{K}(c,c')u_n(c')\right]=V_\alpha\lambda_nu_n(c)$$

Although their eigenvalues for function $\tilde{G}$ have been inflated by multiplicative noise, the eigenvalues estimated by cross-validated PCA are those of G as the noise has mean 0. Thus, $\mathbb{E}\left[\hat{\lambda}_n\right]=\lambda_n$. In addition to these eigenfunctions,

$\tilde{G}$ has a new set of eigenfunctions $b_m(c)$ corresponding to the eigenfunctions of the additive noise covariance B. These are orthogonal to the directions of signal covariance, and thus contribute estimated eigenvalues of 0. Thus, under the assumed noise model, cvPCA provides an asymptotically unbiased estimate of population eigenvalues, together with an additional set of zero eigenvalues resulting from orthogonal additive noise. □

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